Palo Alto Networks Cybersecurity Academy –Setting up a PA220 Firewall for a SOHO Environment

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Purpose

The purpose of this lab is to set up a PA220 for a small-scale SOHO network, which is the primary use case for the device. The lab covers skills such as setting up DHCP services and trust boundaries on a router, which are essential for basic network functions and security.

Background

SOHO, which stands for Small Office/Home Office, is a network type commonly used by individuals or small businesses with less than 10 employees. This network type commonly uses smaller-scale routers, switches, and firewalls compared to their large enterprise counterparts. SOHO networks provide numerous advantages to teams of 1-10 people as they are easier to set up and are more affordable than full-size network equipment. SOHO networks often only have a single router, and may contain switches, wireless access points, and end devices such as computers and printers.

The Palo Alto Networks PA-220 next-generation firewall, contrary to their core product, is designed for small organizations or branch offices and includes the following main features: active/passive and active/active high availability (HA), passive cooling (no fans) to reduce noise and power consumption, eight Ethernet ports, and dual power adapters for power redundancy. The PA-220 firewall enables you to secure your organization through advanced visibility and control of applications, users, and content.   Marketed as a NGFW, Next-generation firewall. The PA220 uses machine learning to identify attacks instead of relying on simple signature check like traditional firewalls. This technology allows the PA220 to identify undocumented threats and brand-new exploits without intervention from Palo Alto networks themselves. The PA220 also prevents threats by filtering URLs and securing against DNS-based attacks.

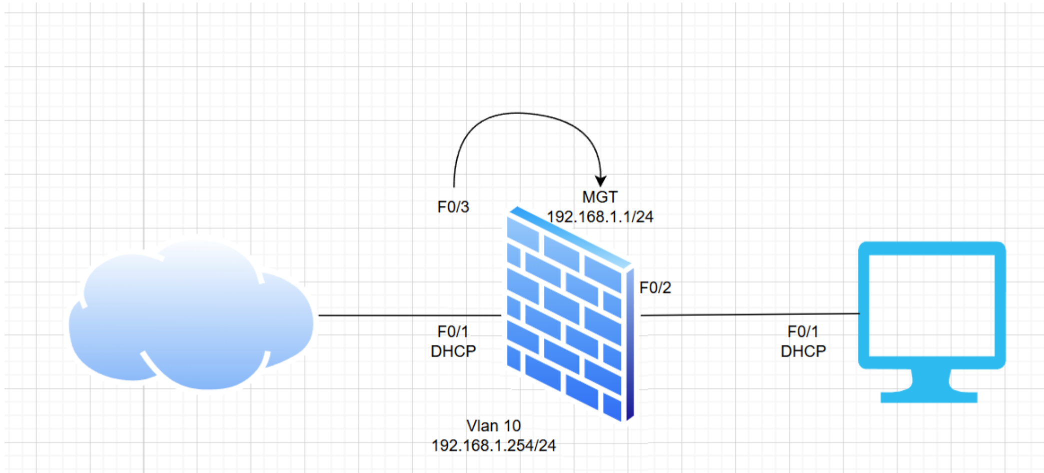
Since the PA220 is a hardware firewall, meaning it's a physical device, it is a more tangible device compared to a virtual software firewall. By using this legacy form factor, the PA220 can provide a higher degree of reliability as the physical device can be troubleshooted by the end user. With a software firewall, security is managed remotely by a 3rd party company, meaning that any issues with the outside company could leave software firewalls users vulnerable for hours or even days. Hardware firewalls put this control in the hands of the users, allowing security to be implemented immediately by an experienced technician.

Palo Alto firewalls run on an operating system called PAN-OS. PAN-OS can be controlled through two methods: a Graphical User Interface (GUI) and a Command-Line Interface (CLI). The GUI is accessible through an HTTP connection and displays in any modern web browser. The HTTP connection is available through the firewall's MGT port and by default, is accessible at https://192.168.1.1. The firewall has a default username and password of admin.

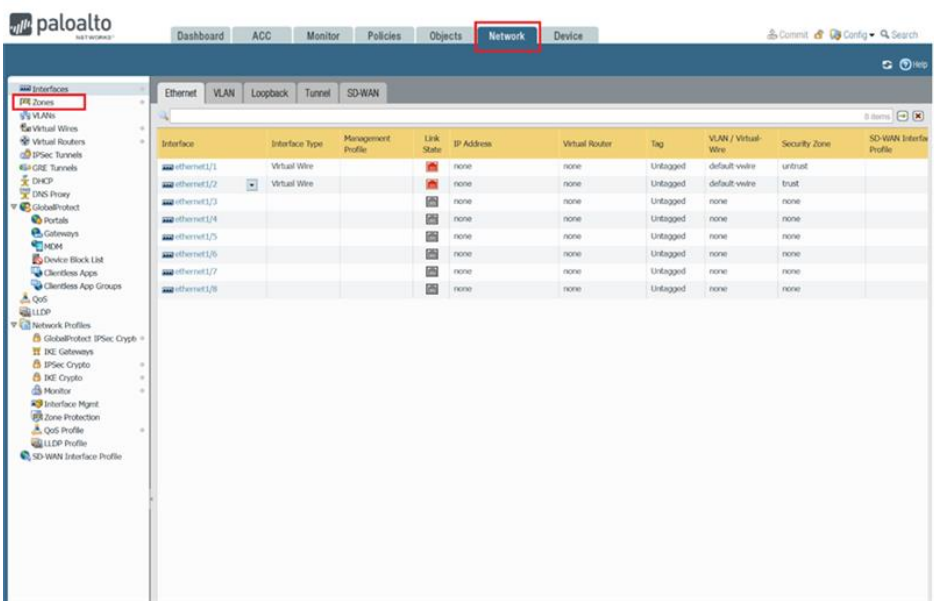
Lab Summary

In this lab, we configured our firewall to connect to a DHCP-enabled ISP router and configured this traffic to be untrusted by default. We then configured the remainder of the router ports to be connected by a single VLAN, have trusted traffic, and be DHCP clients served by the firewall

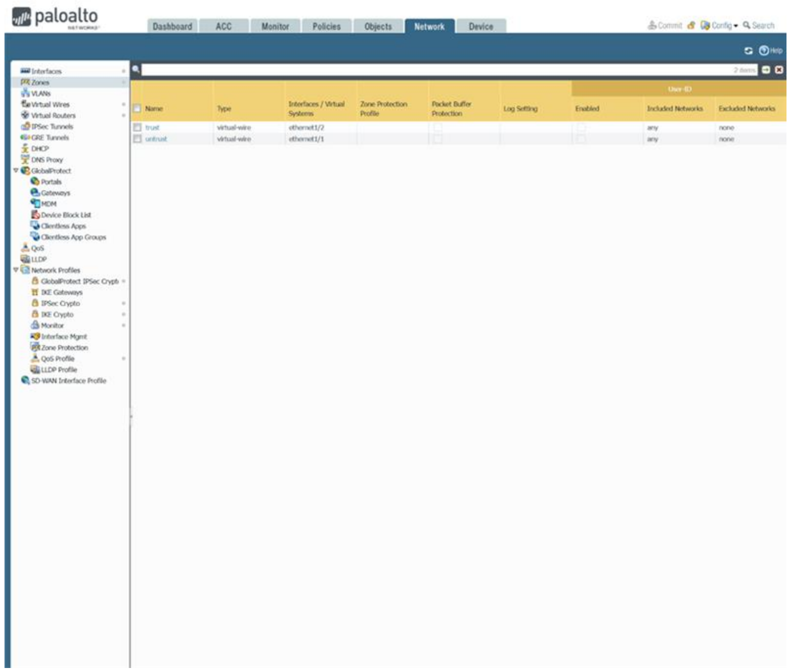
Network Diagrams



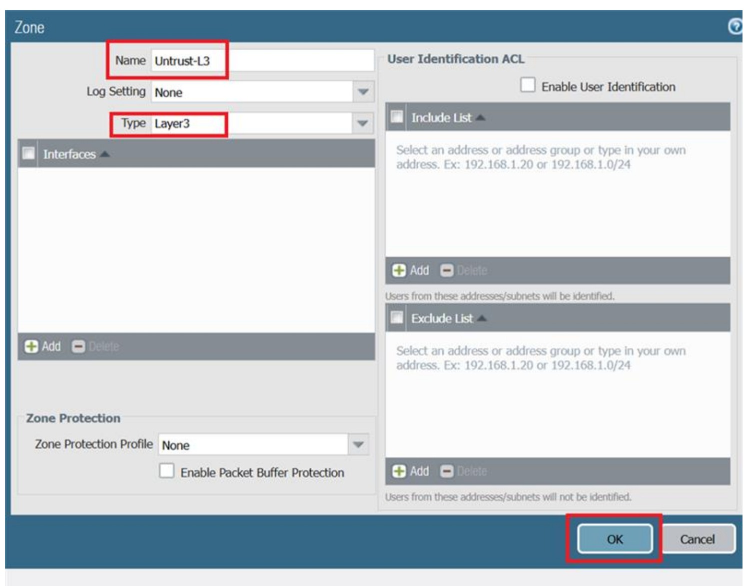
Lab Commands



1. When in your Web GUI go to the Network tab as highlighted in the image above. Then click the Zone section on the left hand side as highlighted in the image above.

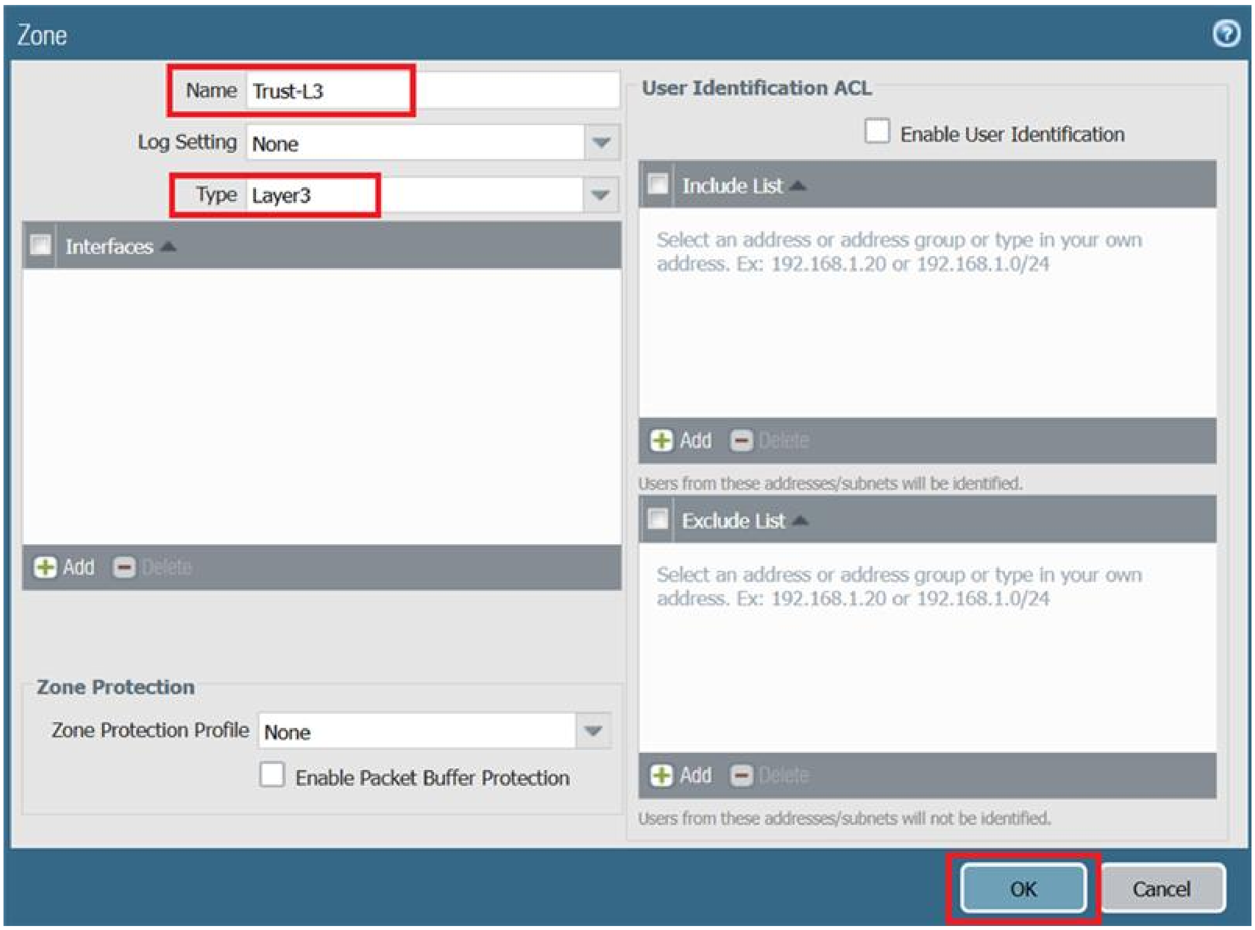


1. Then once on a screen like this click the add button on the bottom of the page.

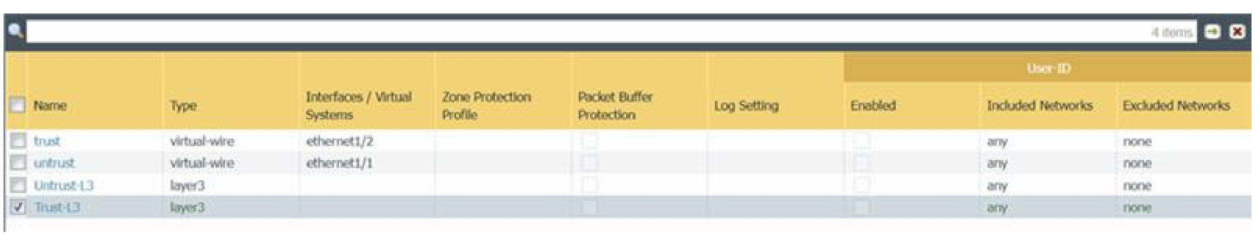


1. Change the Name to Untrust-L3 as highlighted in the image above. Also modify the type to Layer3. Then, Click the OK button as highlighted in the image above.

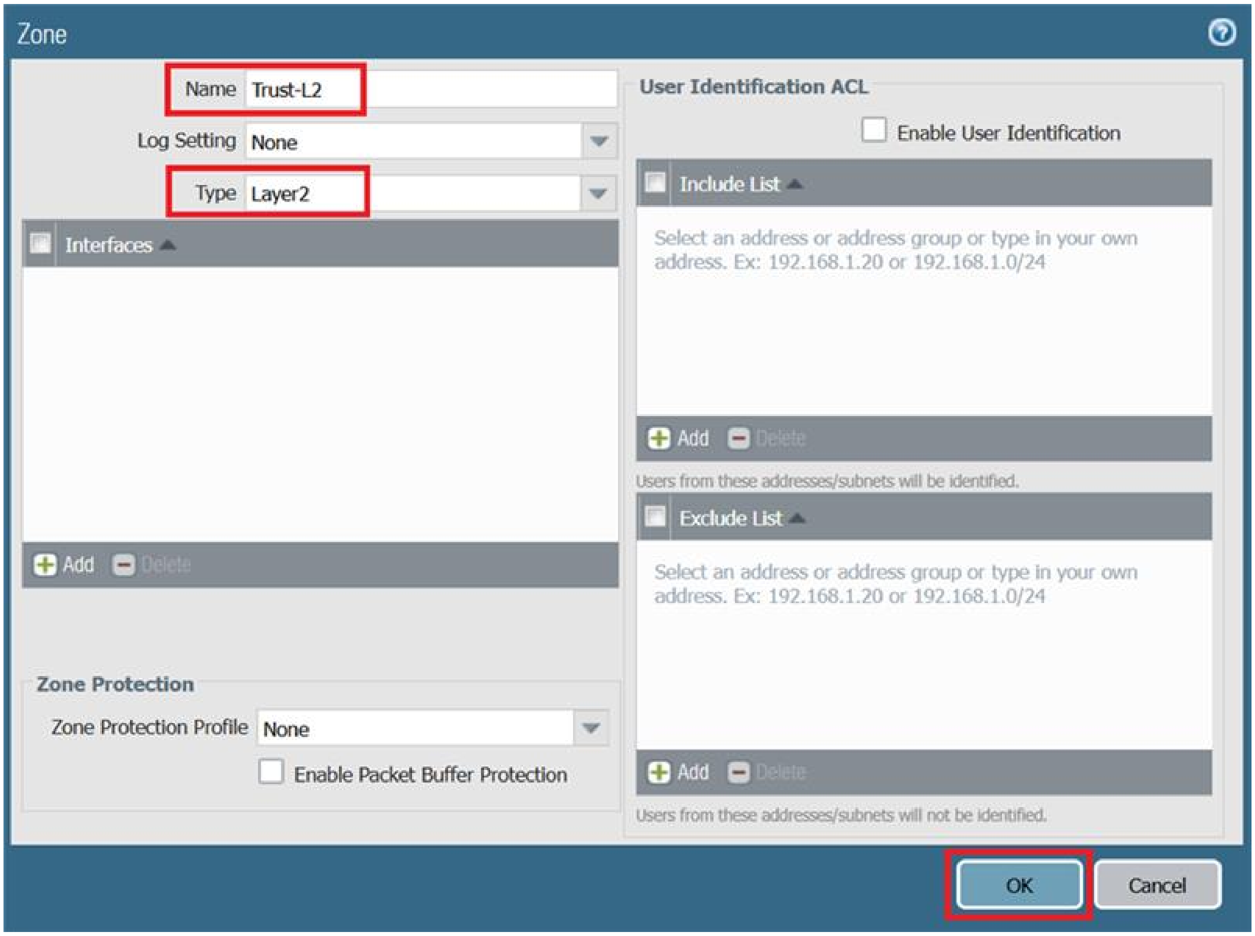
4. Then click the same add button to create another zone.



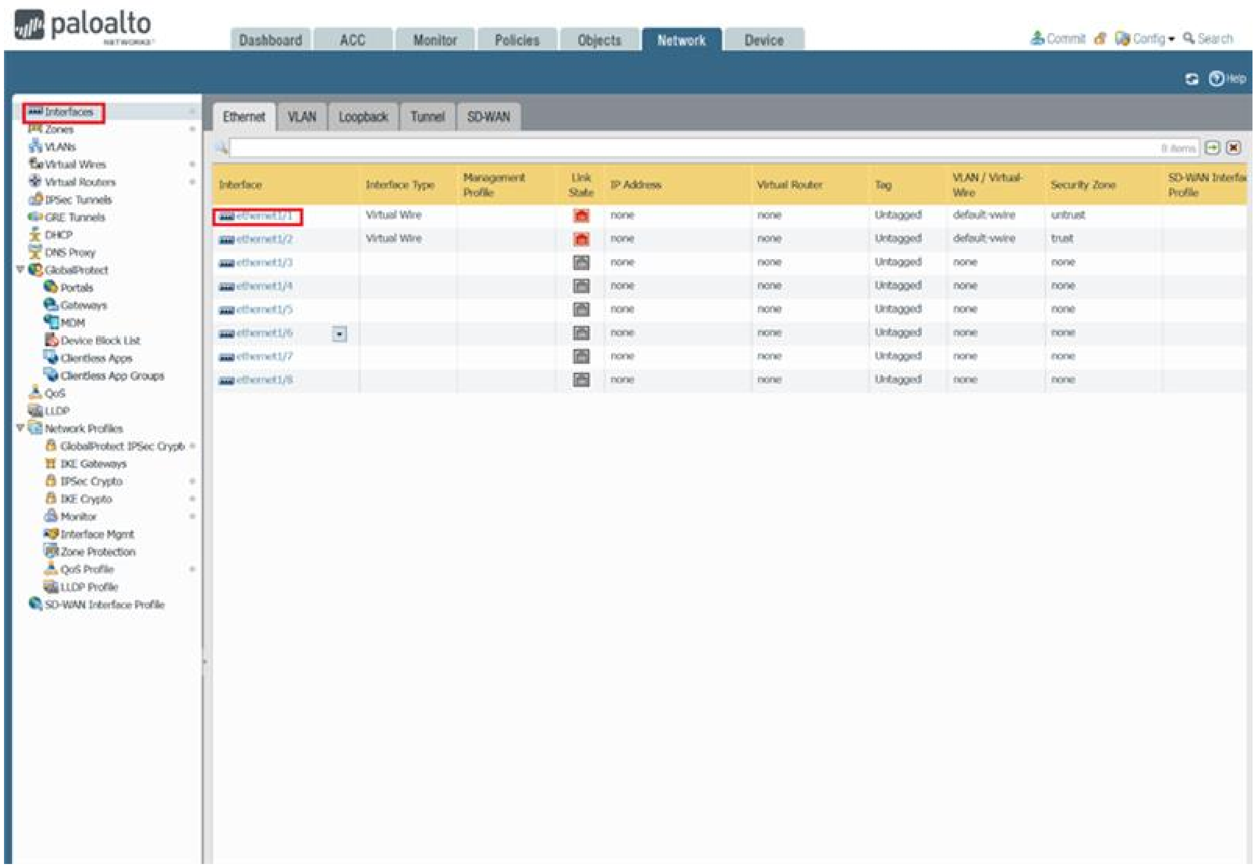
5. Change the Name to Trust-L3 as highlighted in the image above. Also modify the type to Layer3. Then, Click the OK button as highlighted in the image above.



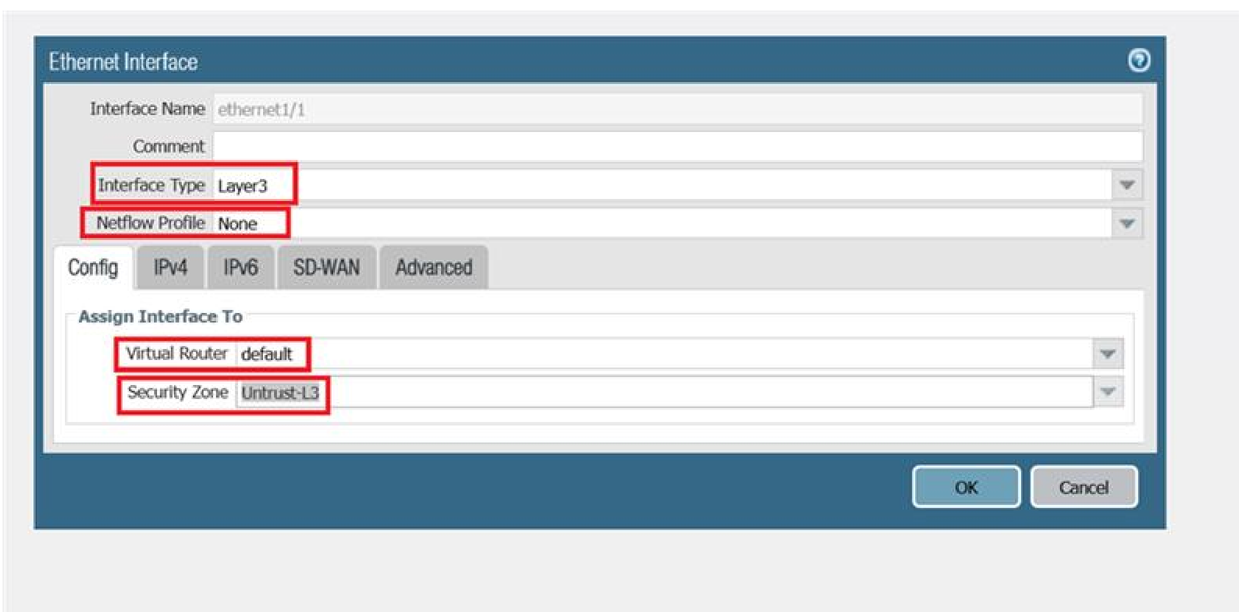
5. Then click the same add zone button to create another zone



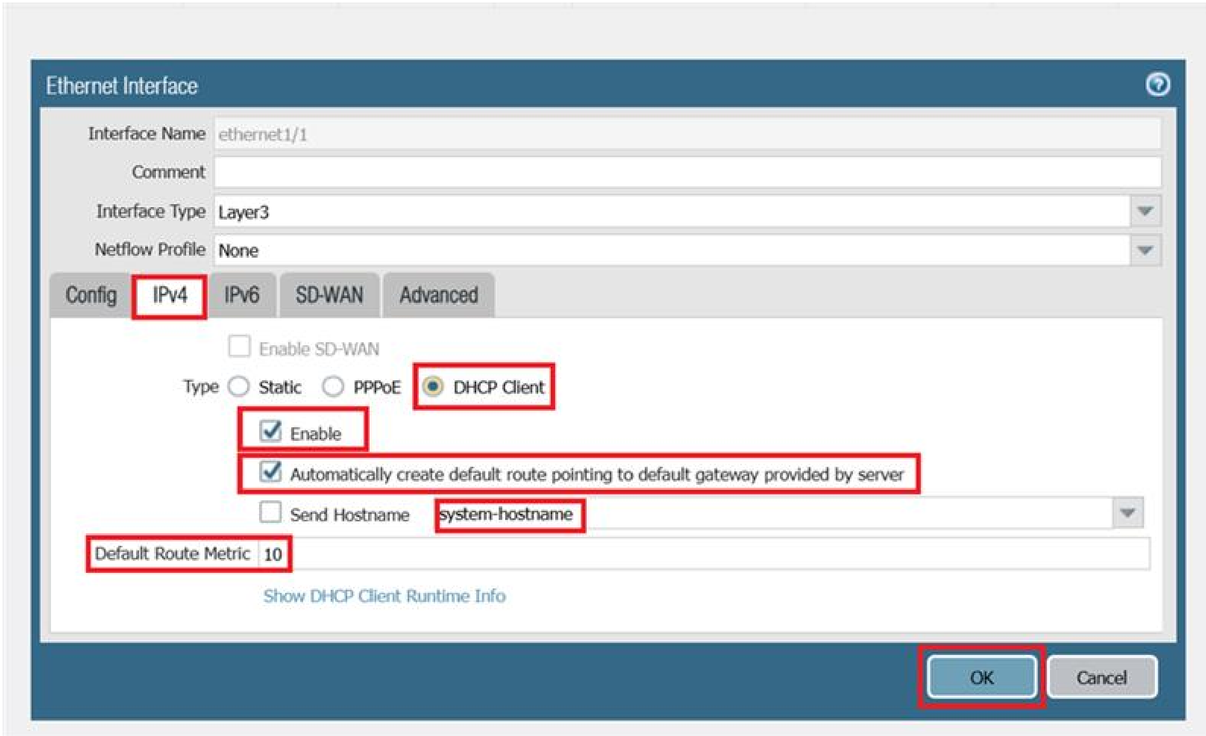
5. Change the Name to Trust-L2 as highlighted in the image above. Also modify the type to Layer2. Then, Click the OK button as highlighted in the image above.



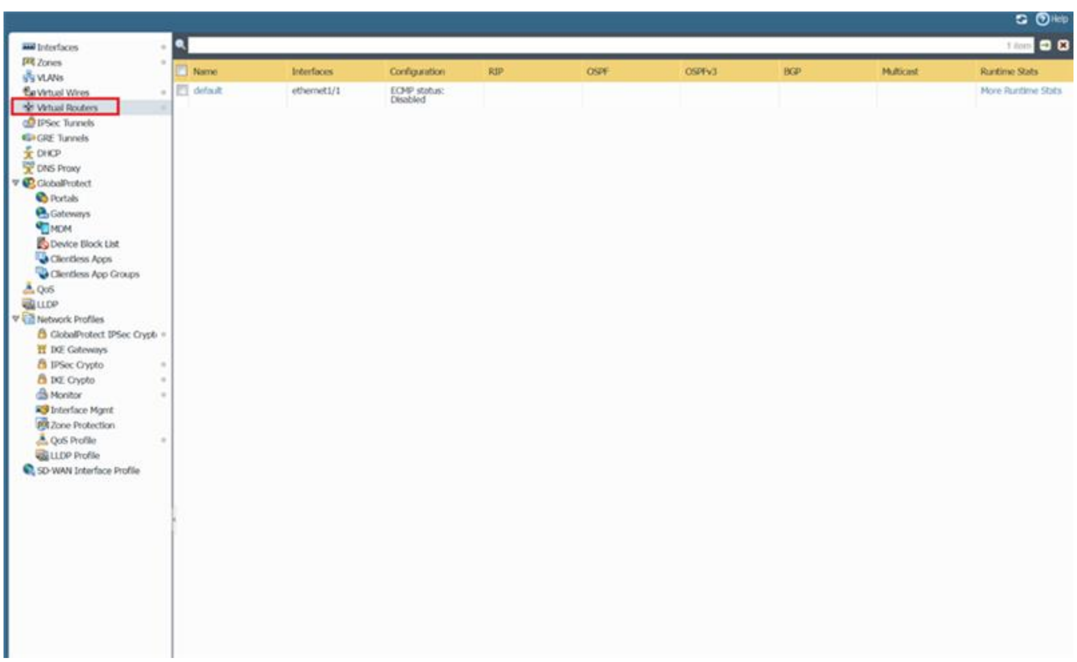
5. Click the Interfaces tab on the left hand side as highlighted in the image above. Then select the Ethernet1/1 option as highlighted in the image above.



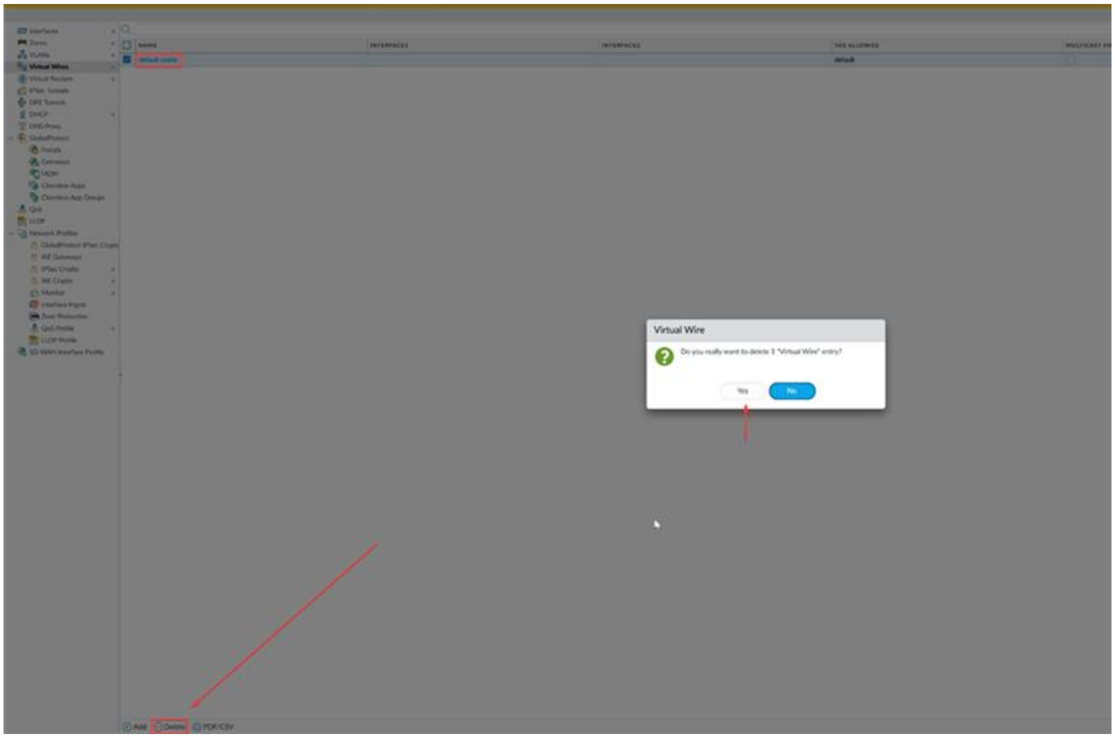
5. Configure the settings as highlighted in the image above.



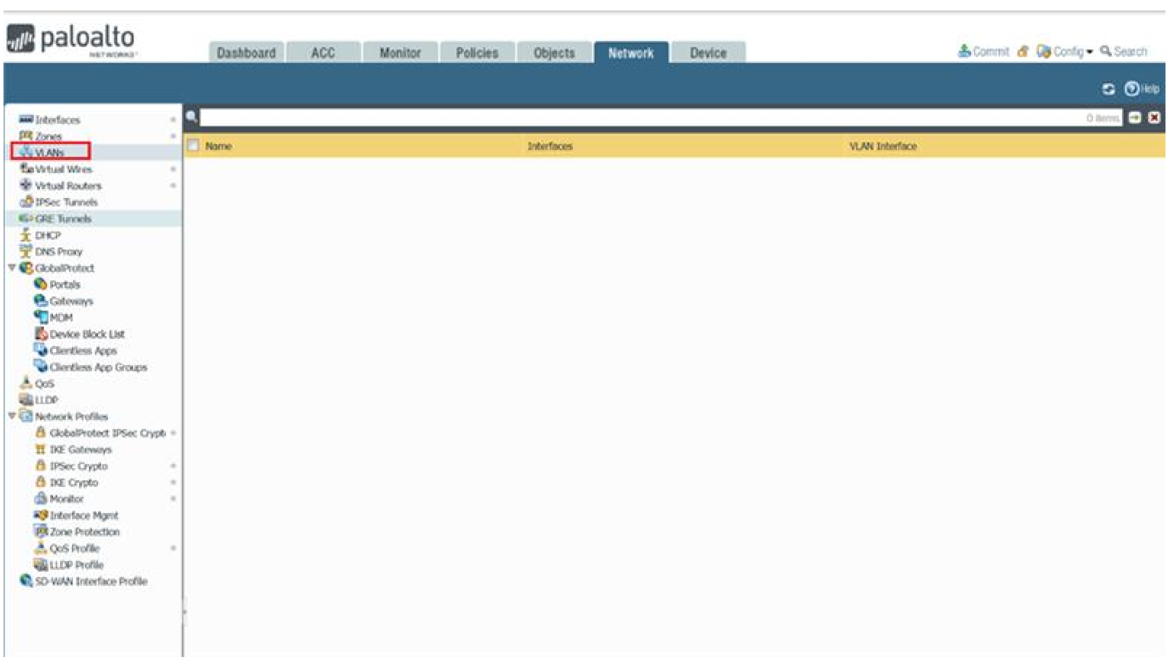
5. Click the IPV4 tab and modify the settings as highlighted in the image above. Then press the OK button as highlighted in the image above.



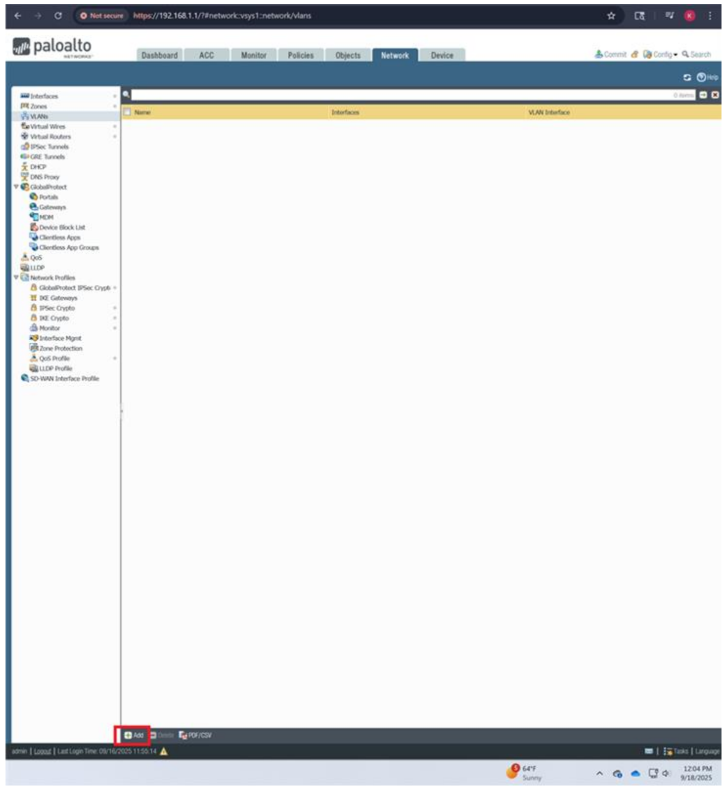
5. Click the virtual interface tab on the left side as highlighted in the image above.



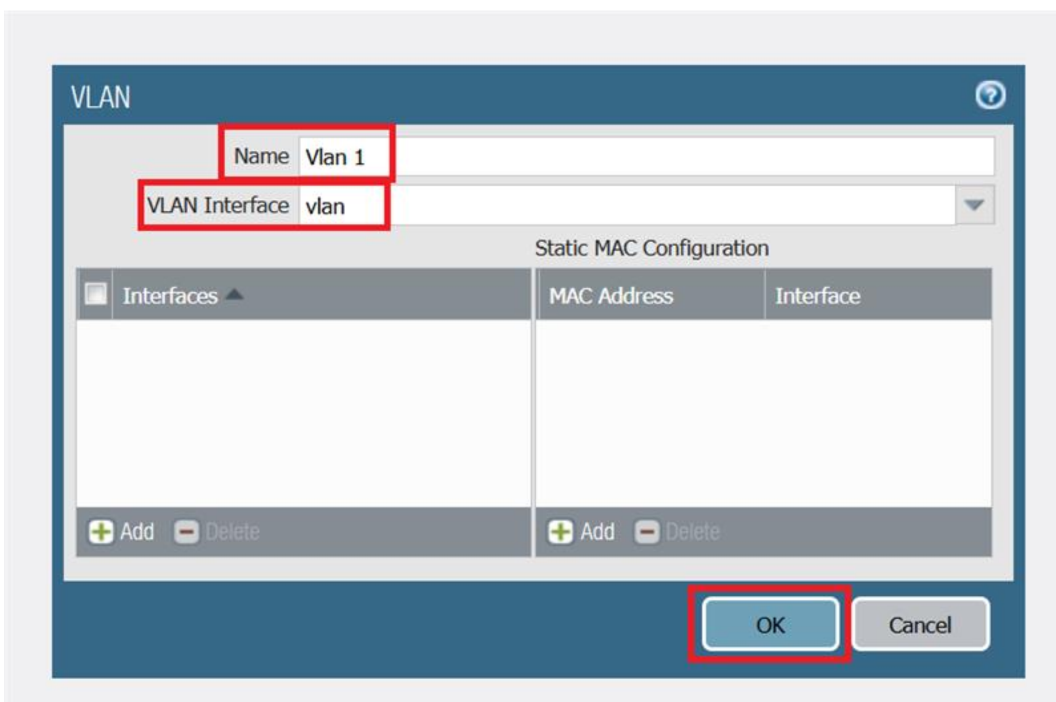
5. Click on the delete button on the bottom of the GUI, select “default”, and then confirm the delete selection.



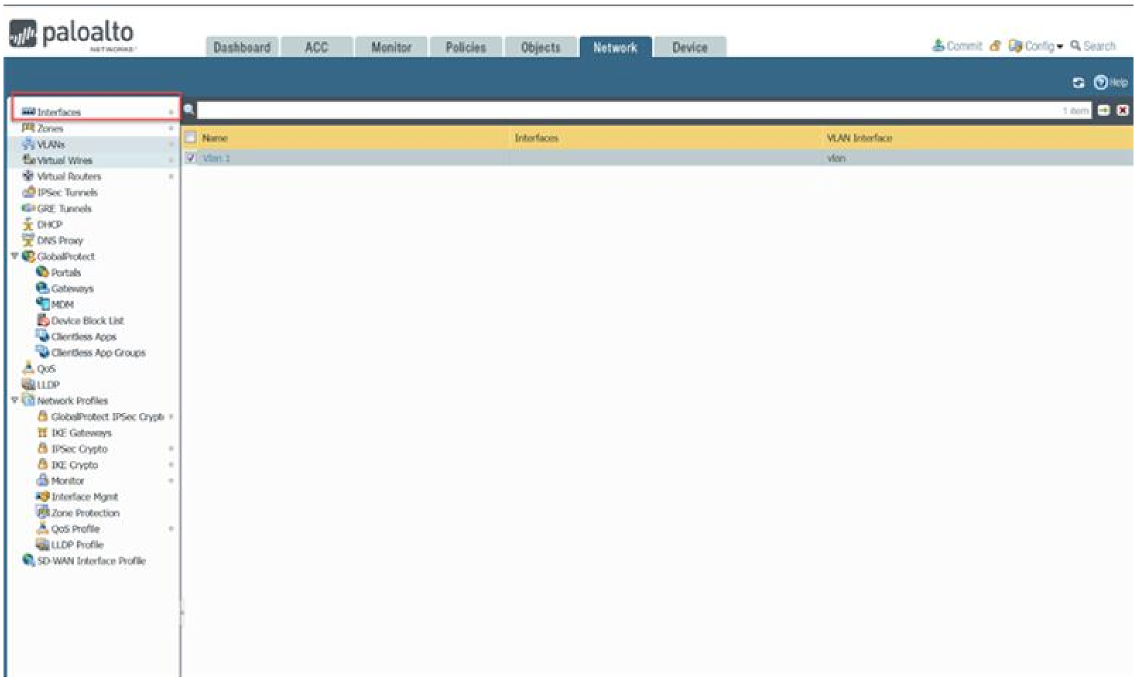
5. Click the VLANs tab on the left as highlighted in the image above.



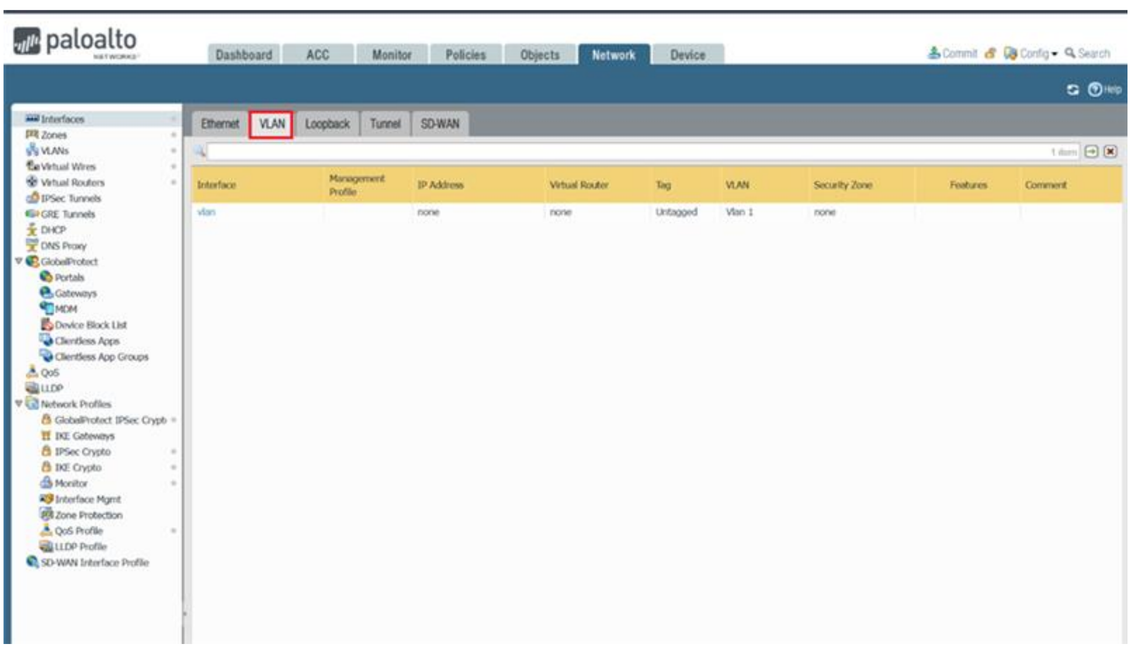
5. Click the add button on the bottom of the GUI as highlighted in the image above.



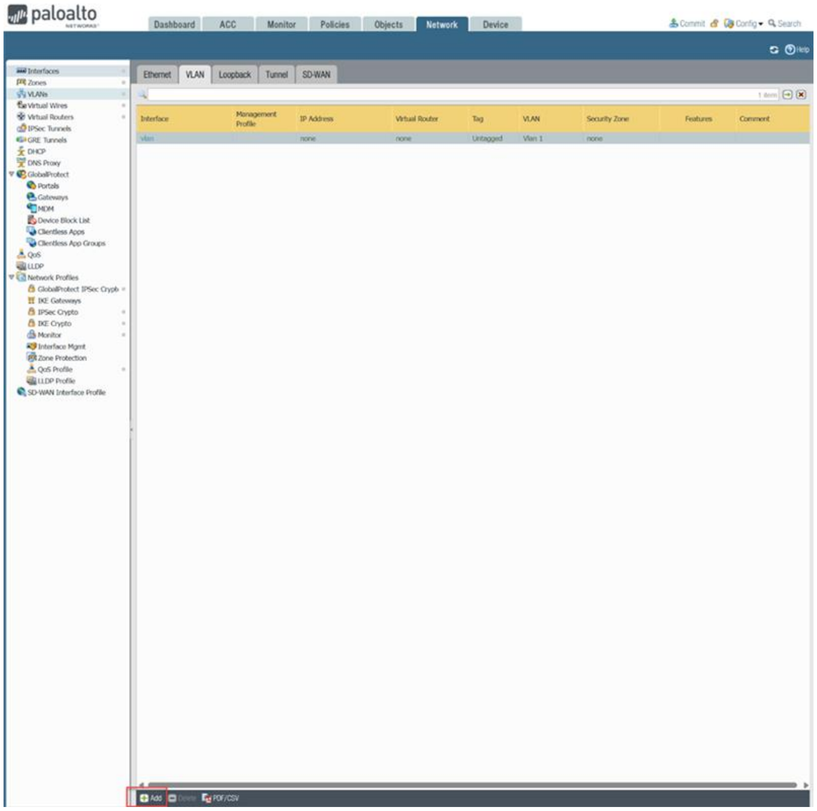
5. Modify the VLAN with the settings as highlighted in the image above. Then, select the OK button.



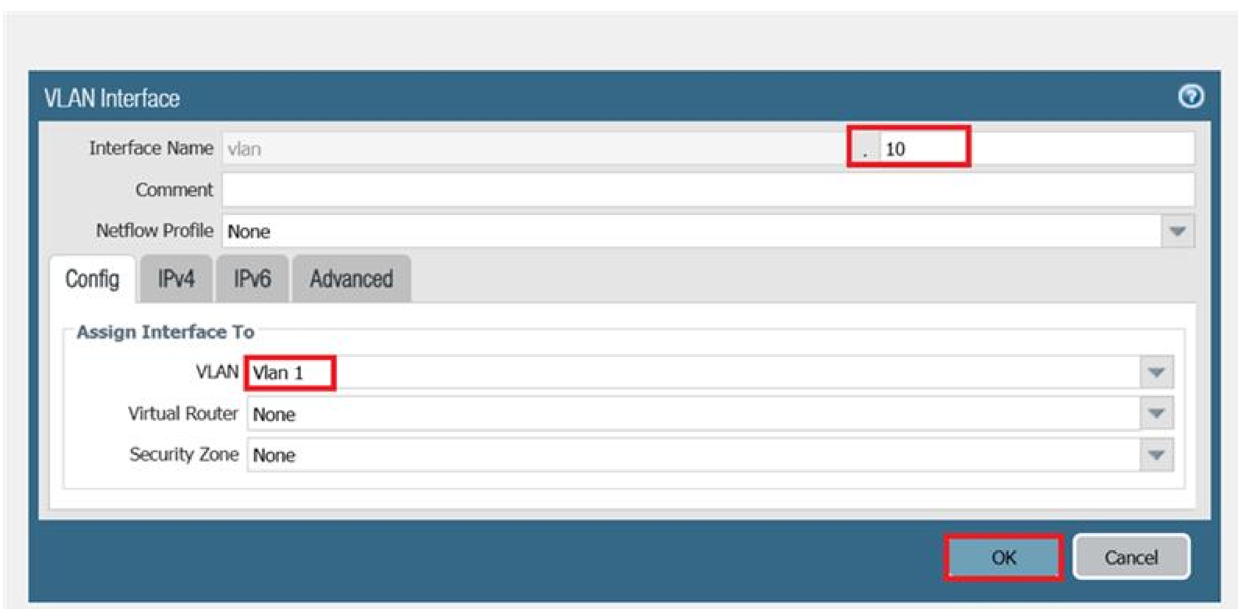
5. Click the Interfaces tab on the left as highlighted in the image above.



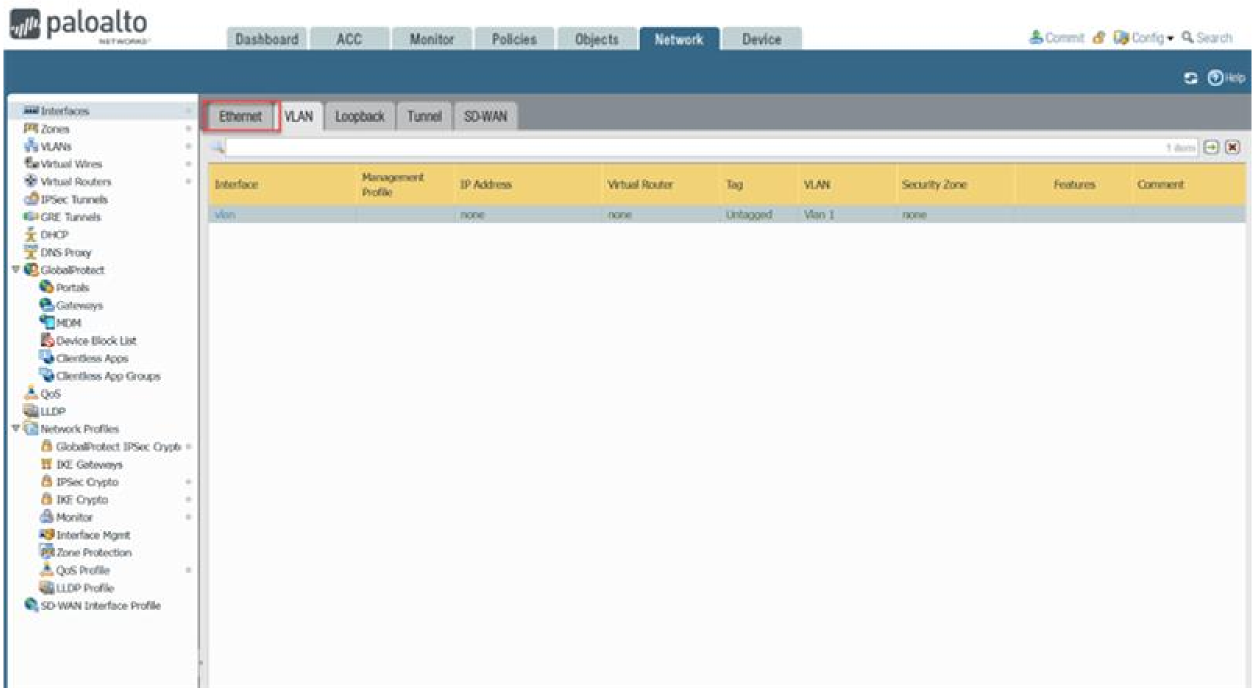
5. Click the VLAN tab as highlighted in the image above.



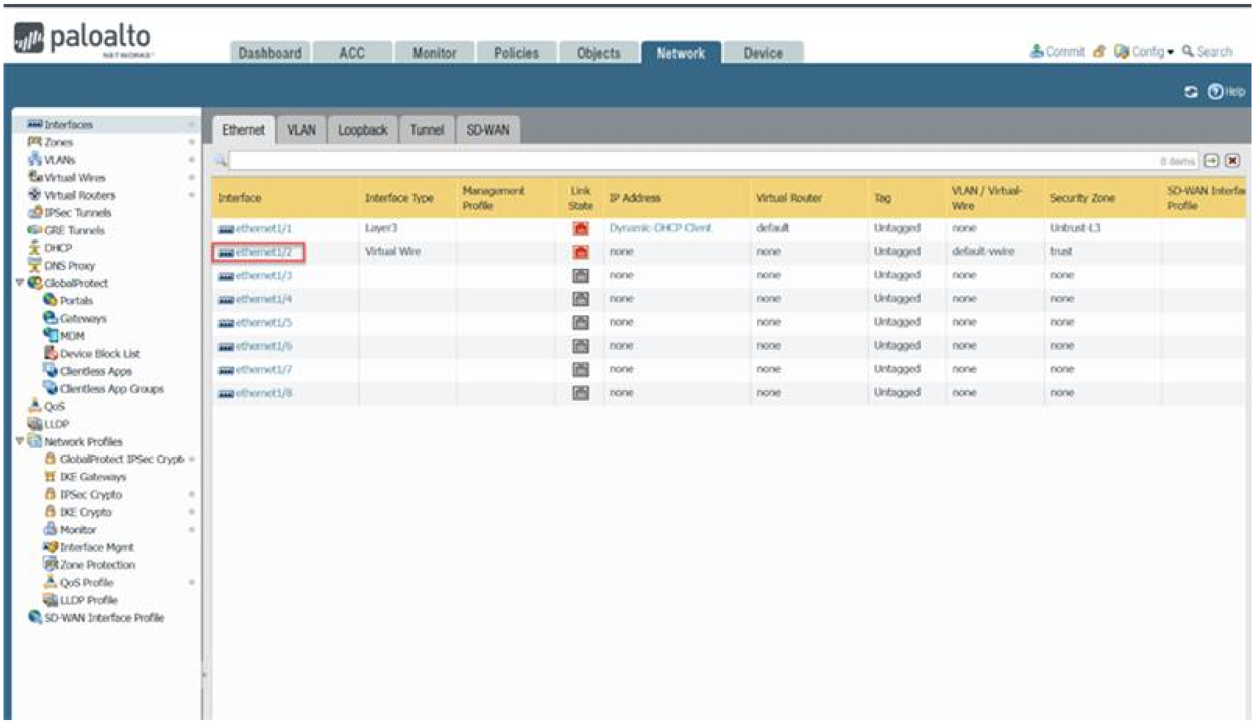
5. Click the add button as highlighted in the image above.



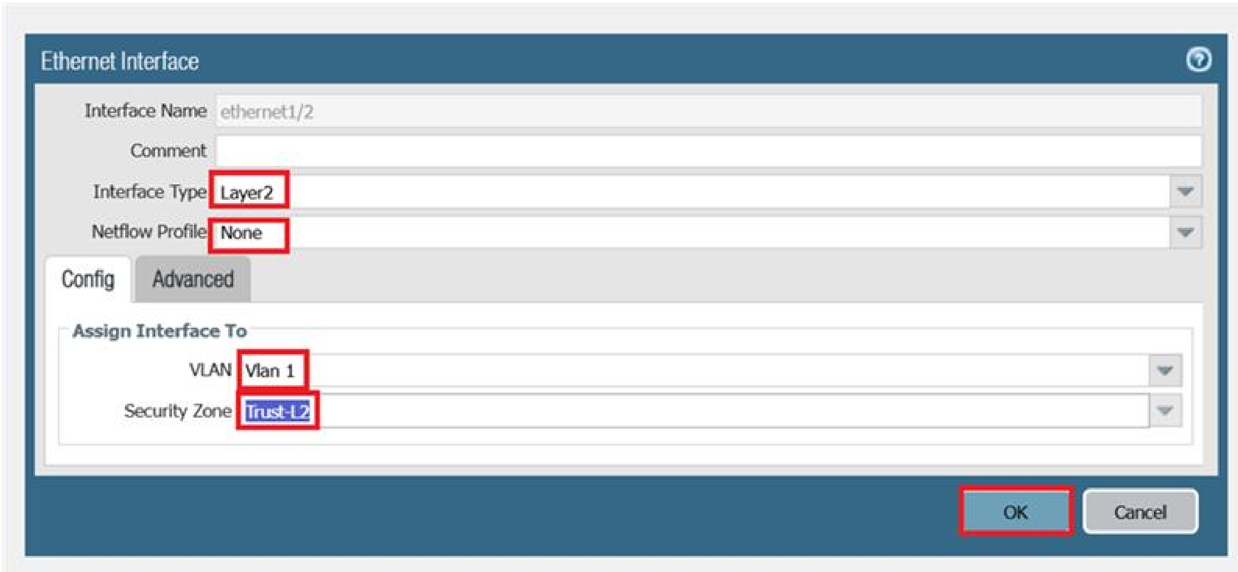
5. Modify the settings as highlighted in the image above.



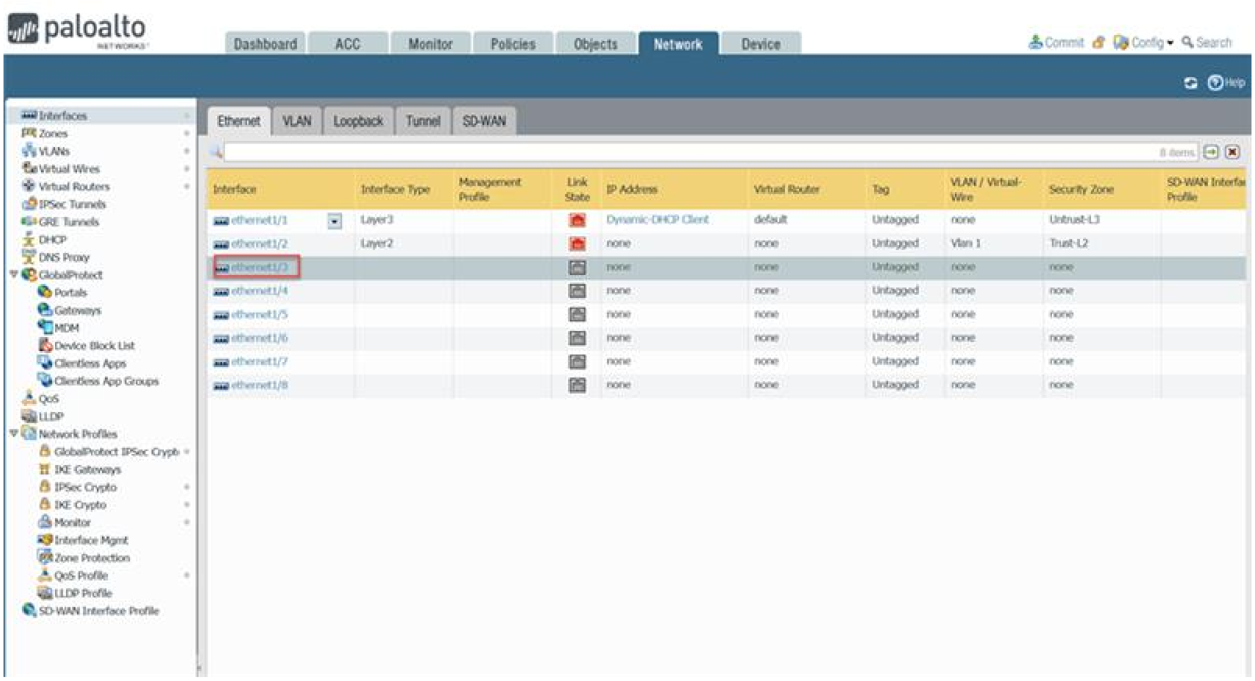
5. Select the Ethernet tab as highlighted in the image above.



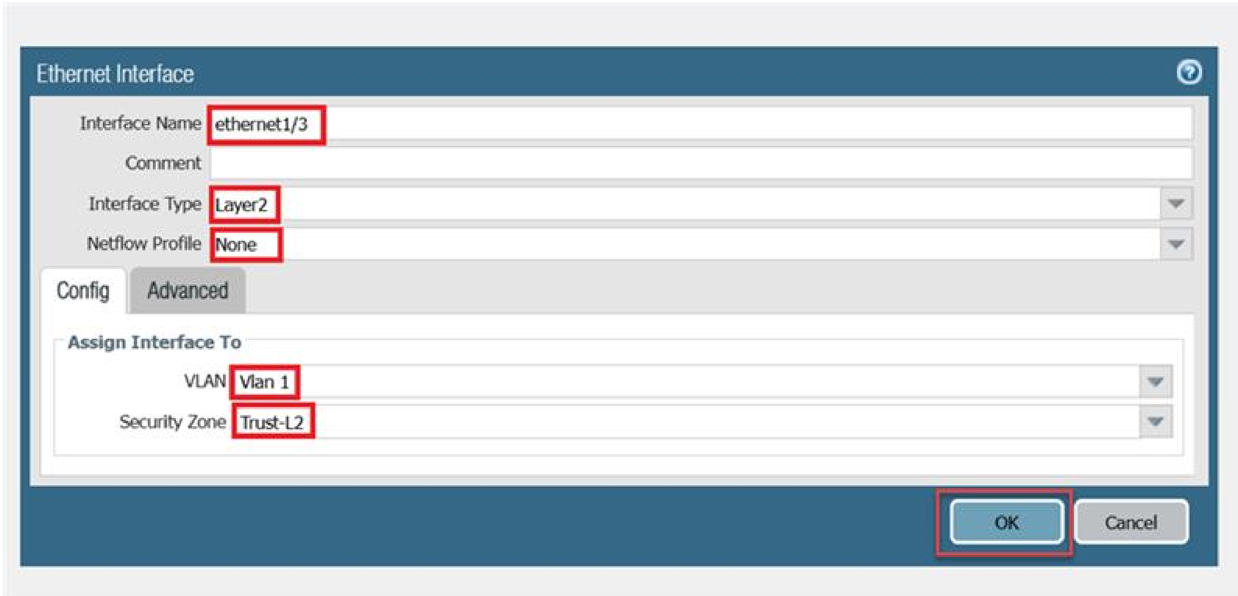
5. Click the Ethernet1/2 as highlighted in the image above.



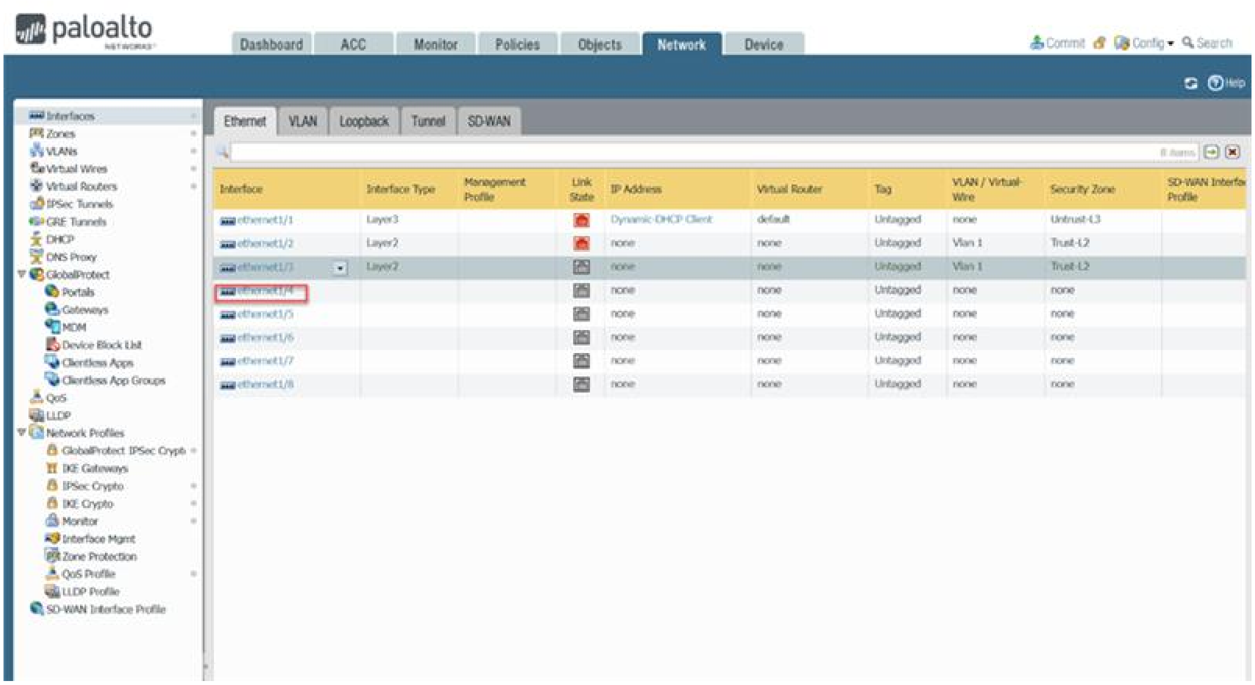
5. Modify the settings as highlighted in the image above. Then, select the OK button.



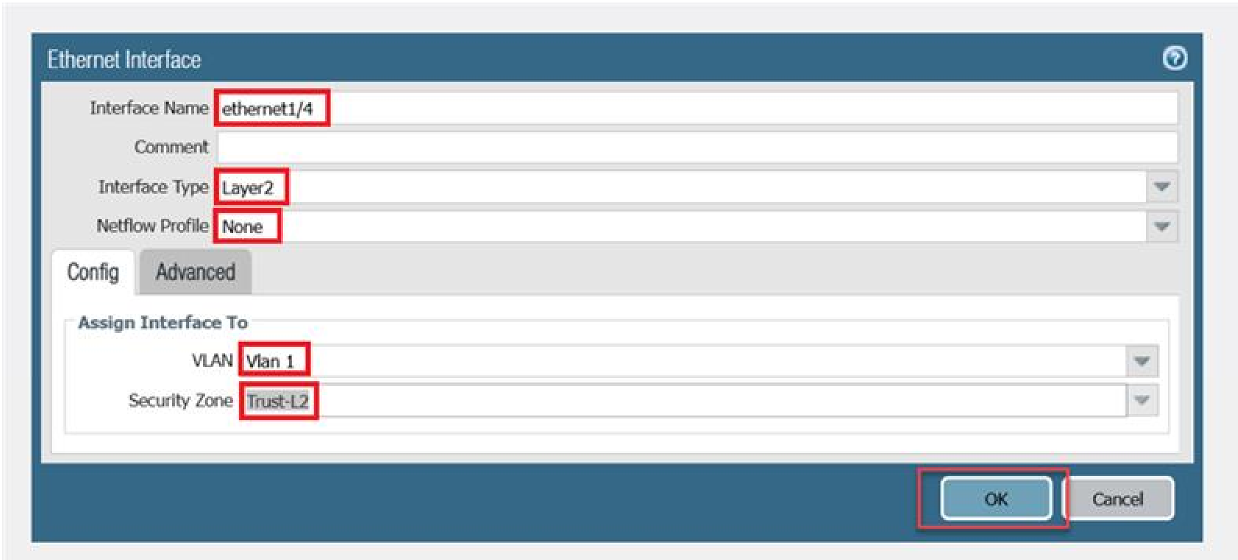
5. Select the Ethernet1/3 as highlighted in the image above.



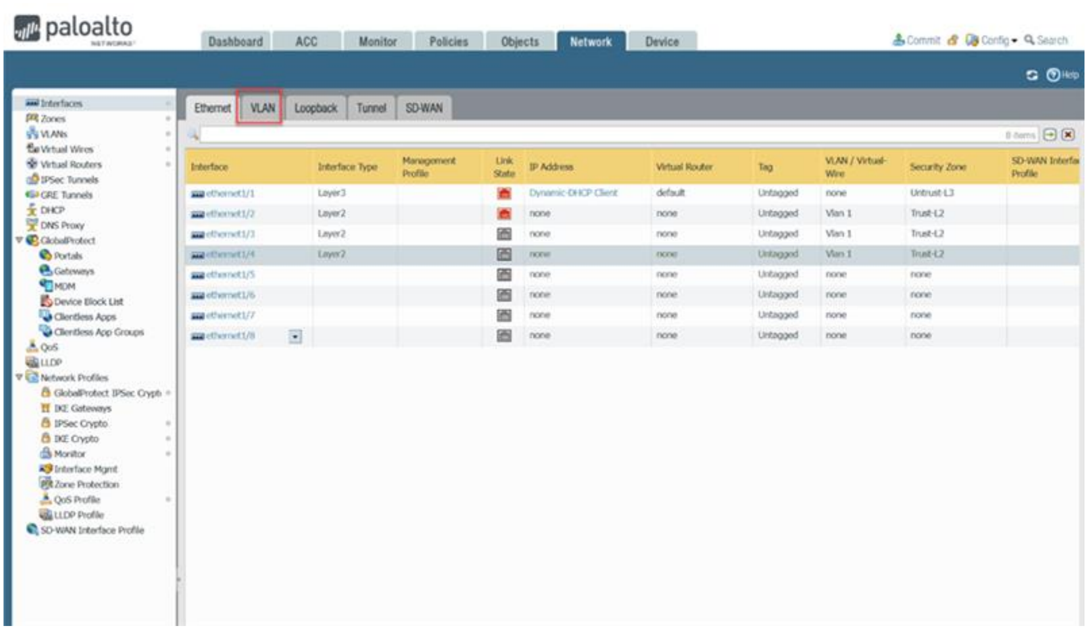
5. Modify the settings as highlighted in the image above.



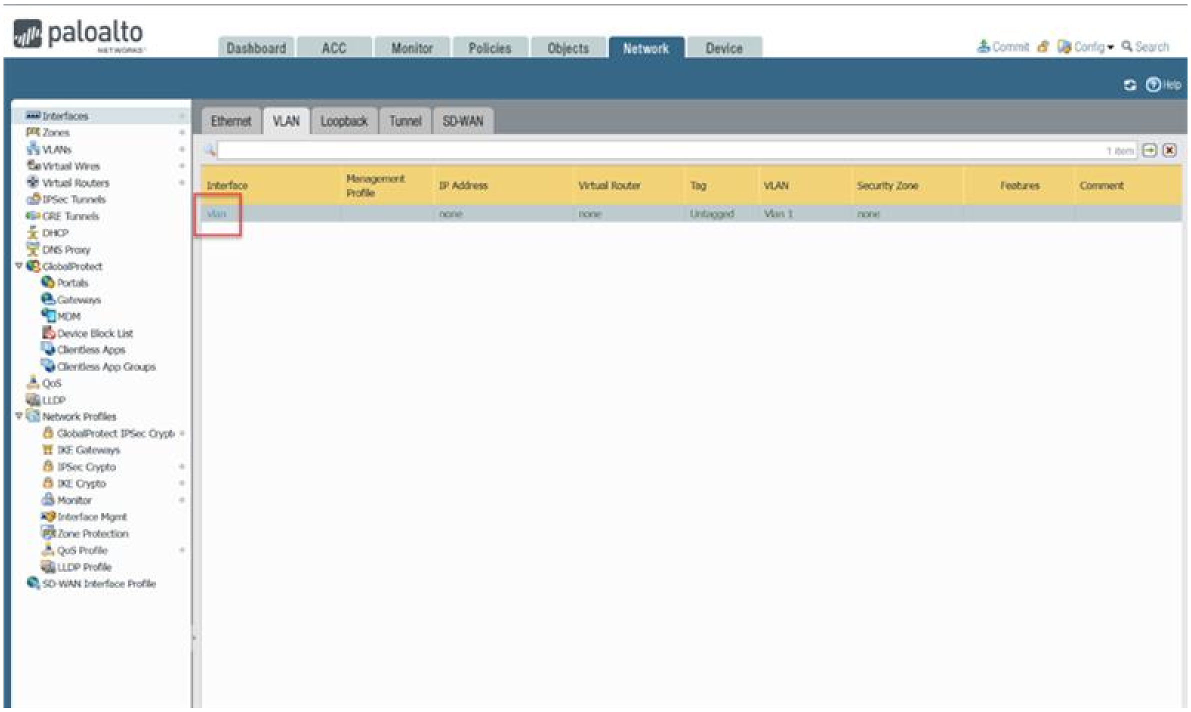
5. Select Ethernet1/4 as highlighted in the image above.



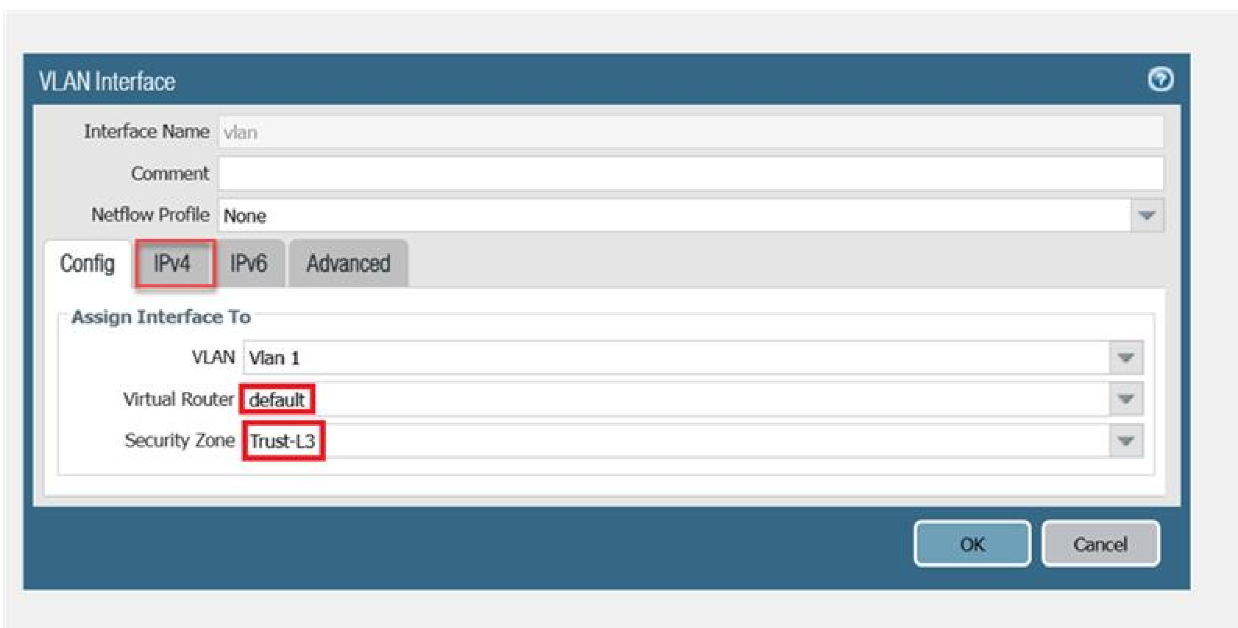
5. Modify the settings as highlighted in the image above.



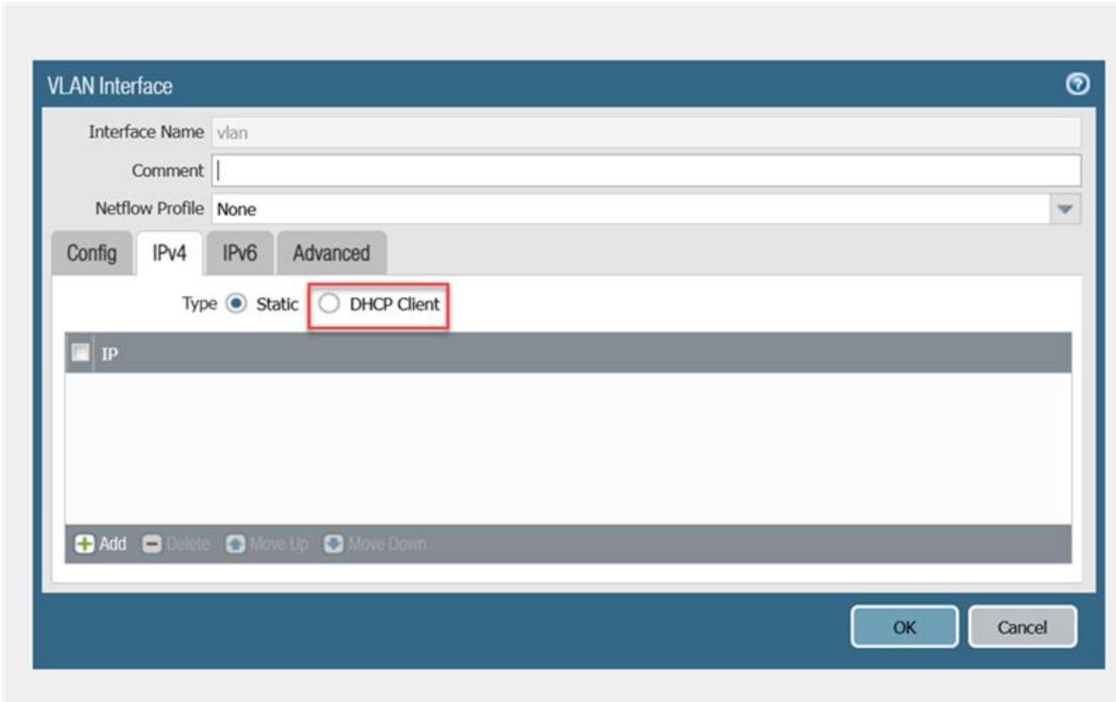
5. Select the VLAN tab as highlighted in the image above.



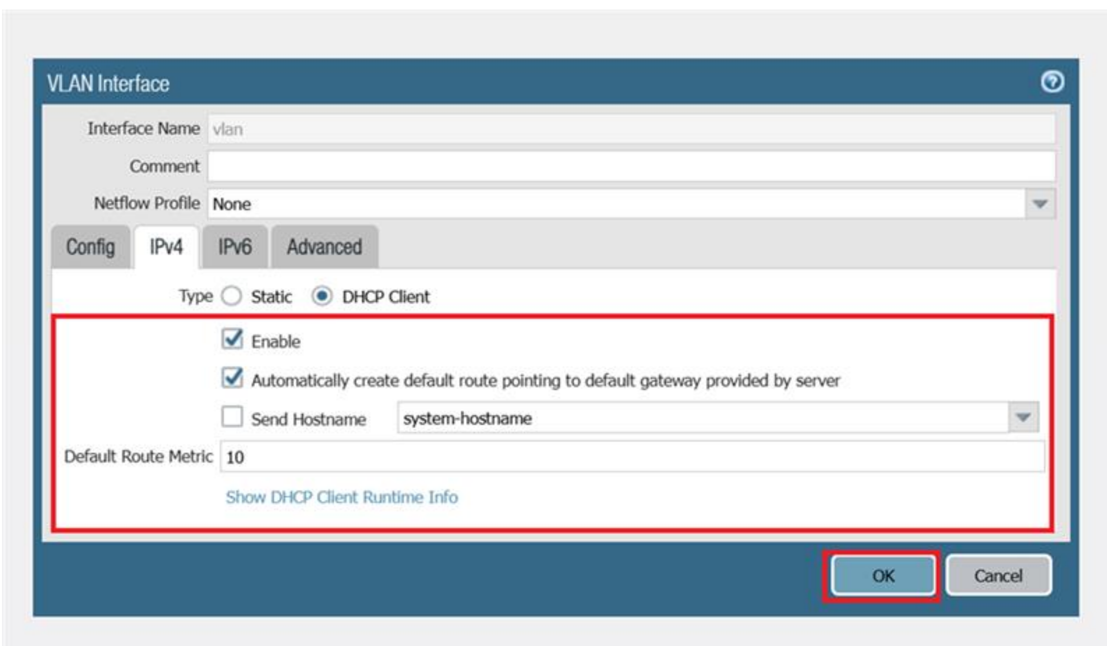
5. Select “vlan” not “vlan.10” as highlighted in the image above.



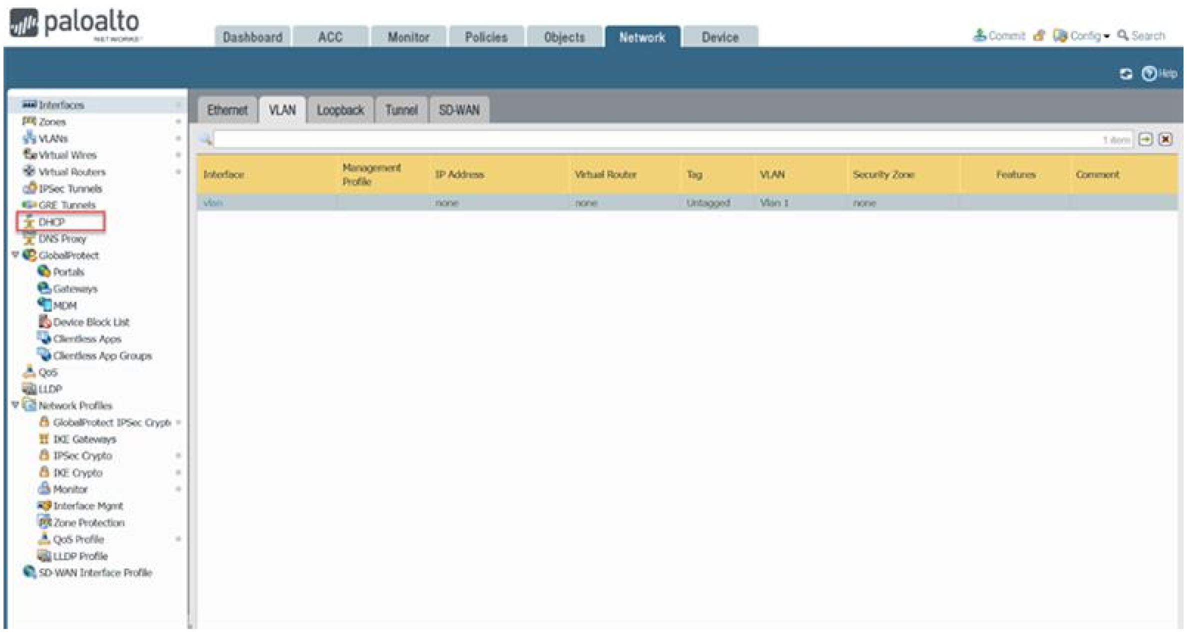
5. Modify the settings as highlighted in the image above. Then Select the IPv4 tab as highlighted in the image above.



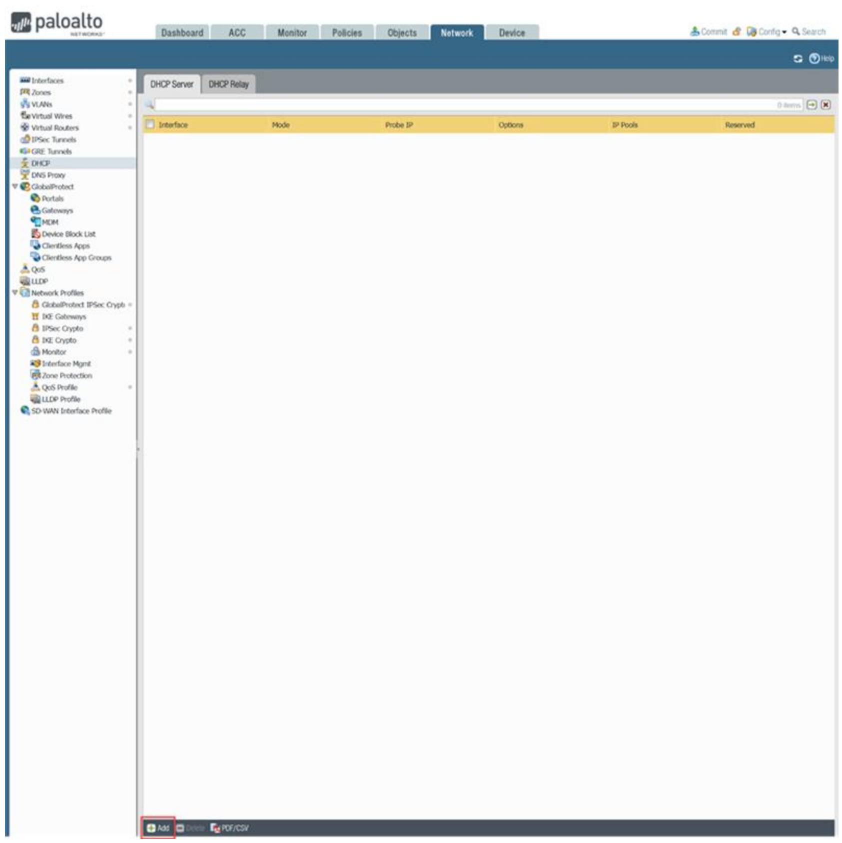
5. Click the DHCP Client option as highlighted in the image above.



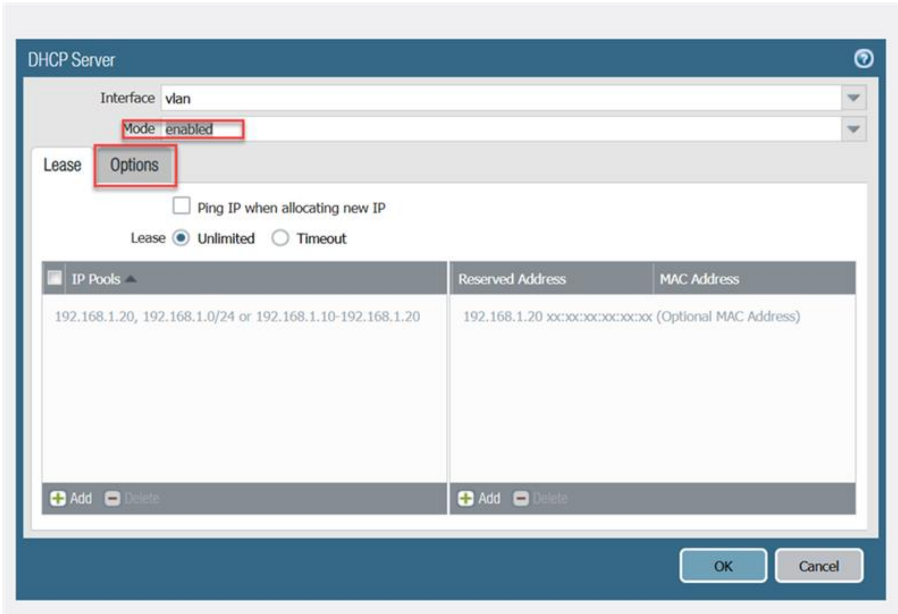
5. Modify the settings as highlighted in the image above. Then select the OK button as highlighted in the image above.



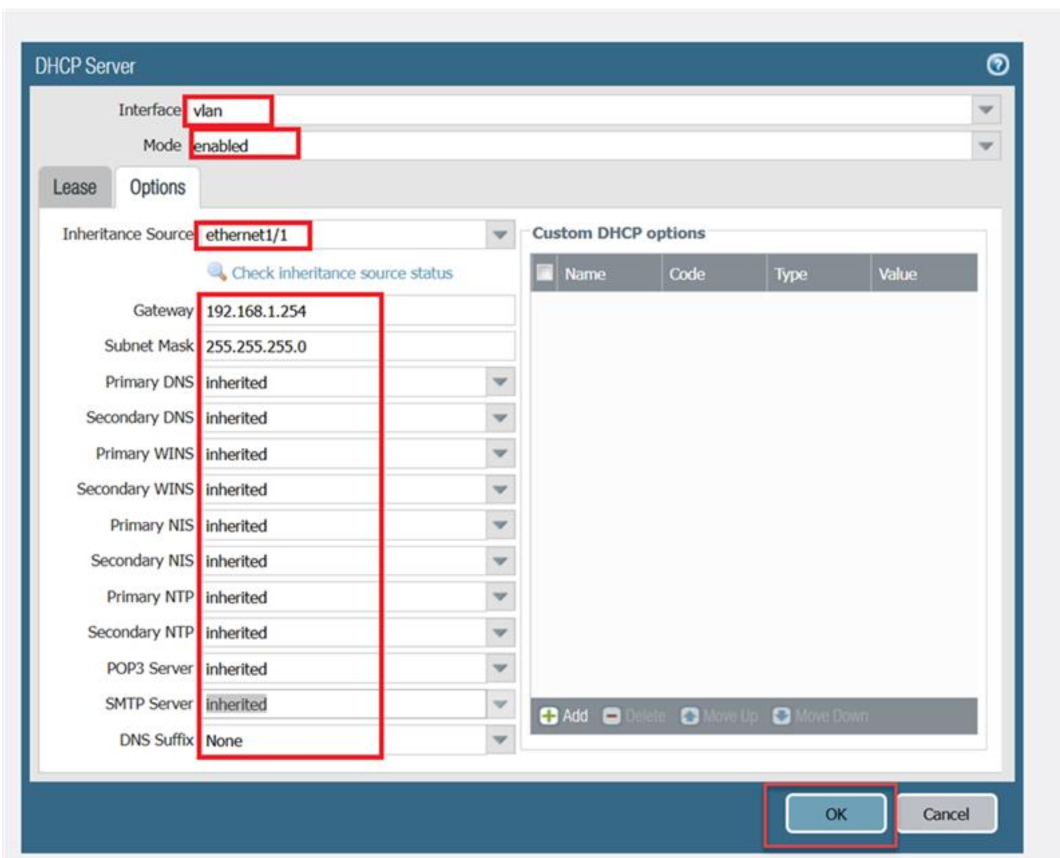
5. Select the DHCP section on the left side of the GUI as highlighted in the image above.



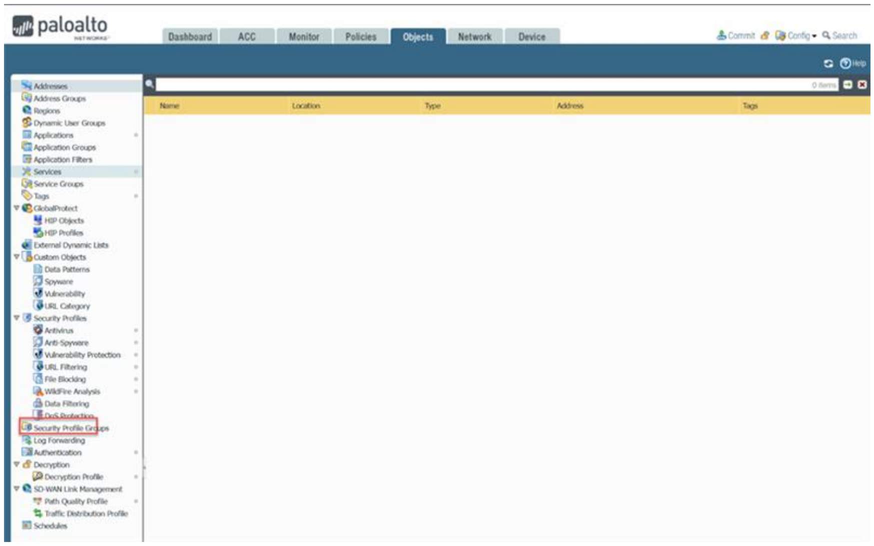
5. Select the Add button on the bottom of the GUI as highlighted in the image above.



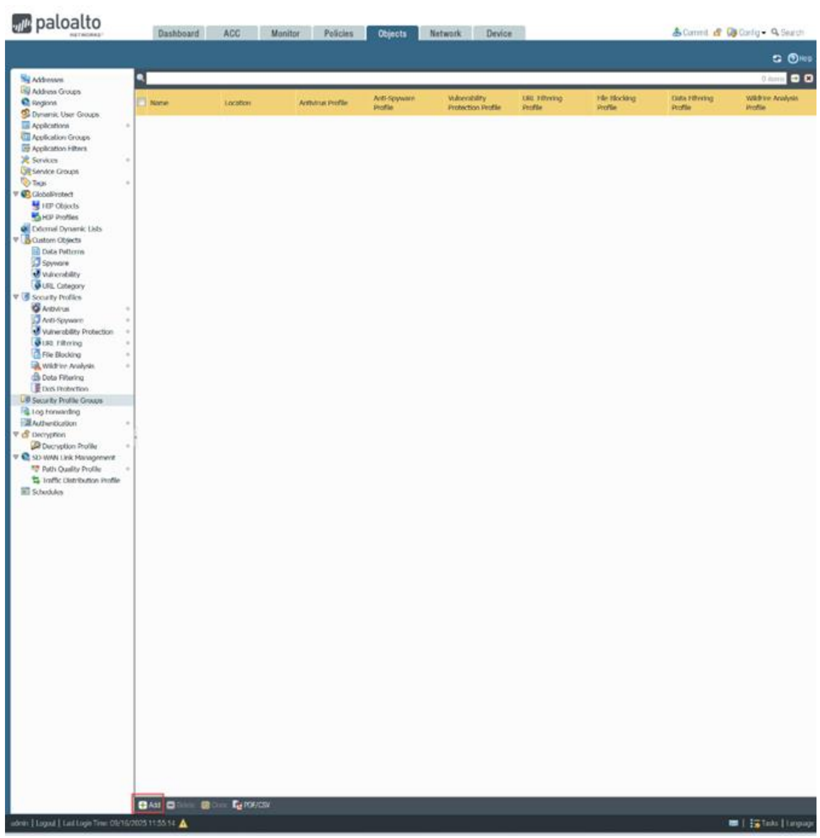
5. Modify the settings as highlighted in the image above. Then, select the Options tab as highlighted in the image above.



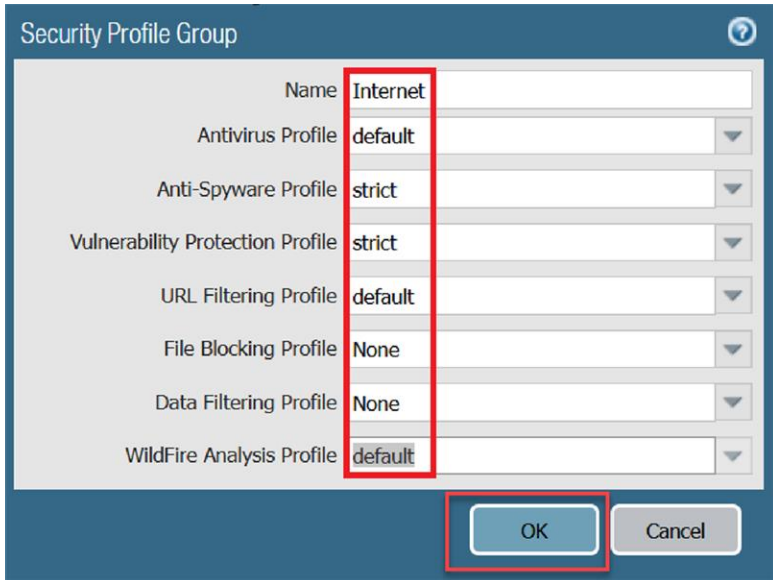
5. Modify the settings as highlighted in the image above, except Change DNS Suffix to "inherited." Then select the OK button.



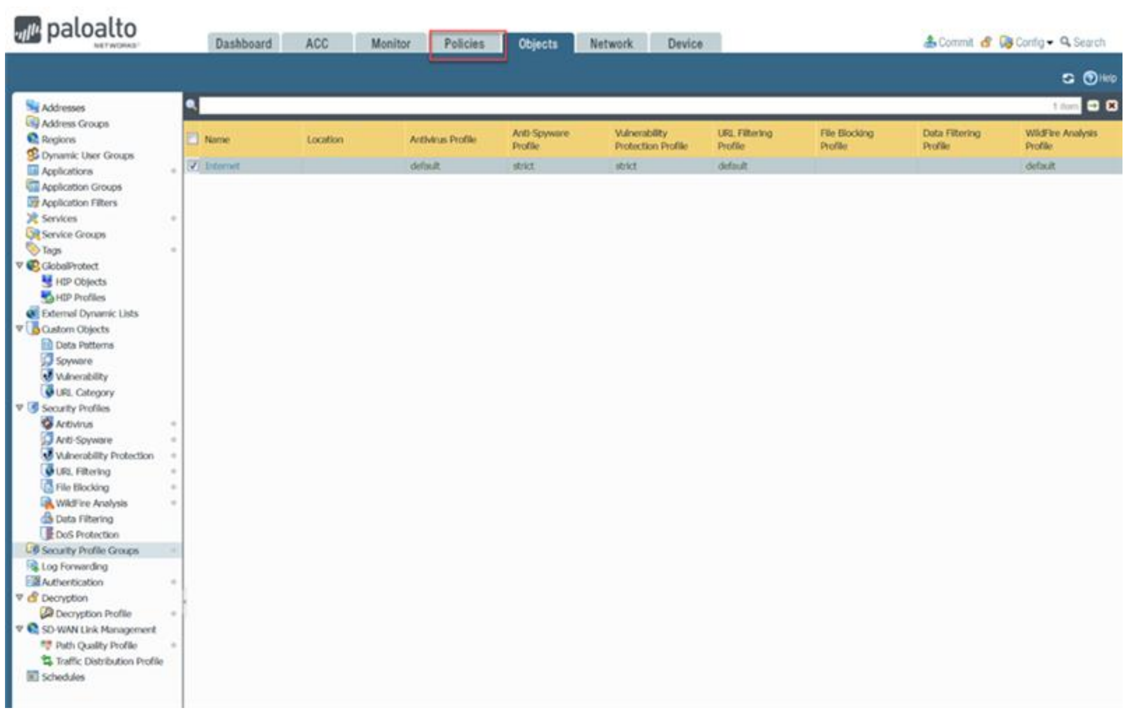
5. Select the Security Profile Groups section on the left side of the Web GUI as highlighted in the image above.



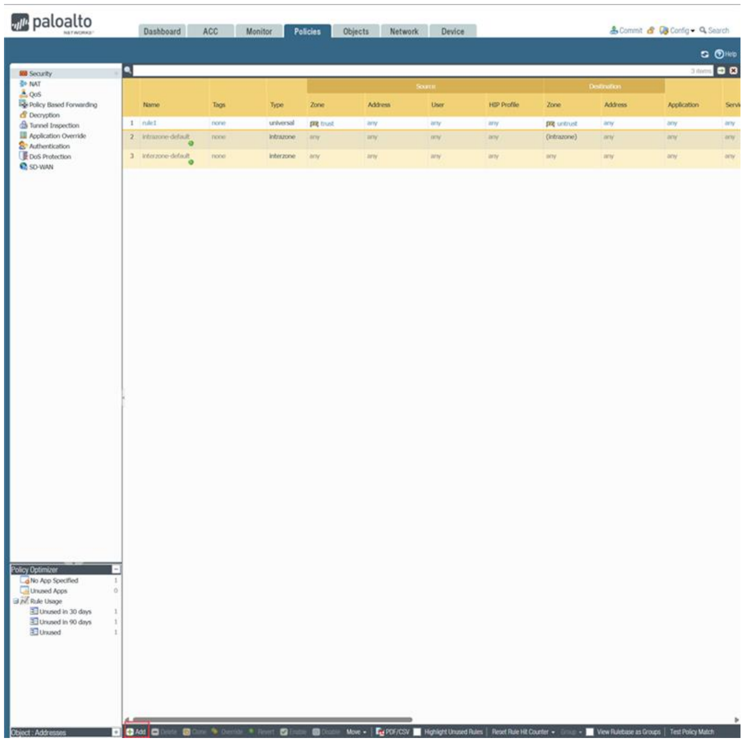
5. Click the Add button on the bottom of the GUI as highlighted in the image above.



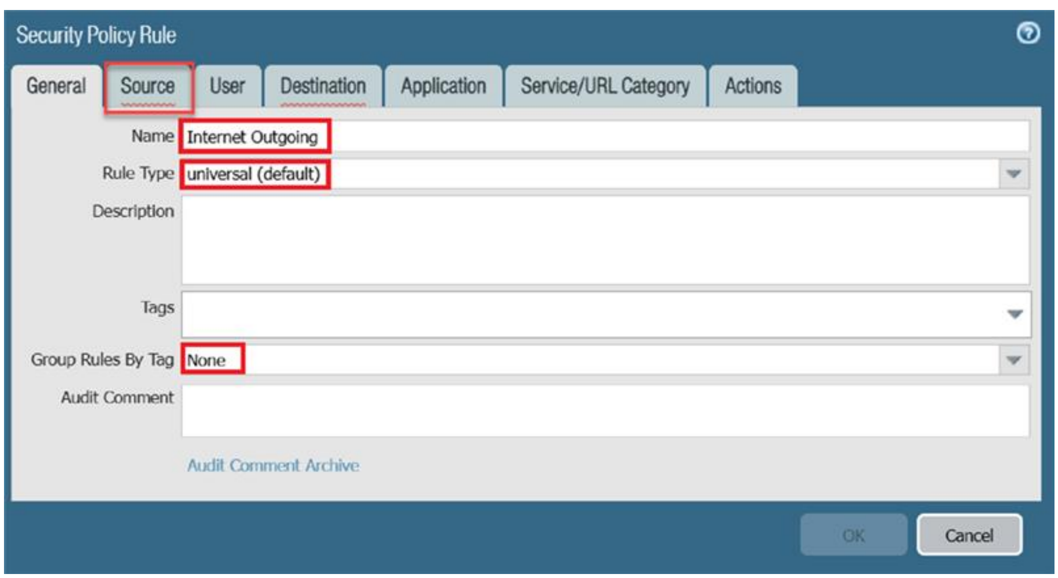
5. Modify the settings as highlighted in the image above. Then, select the OK button.



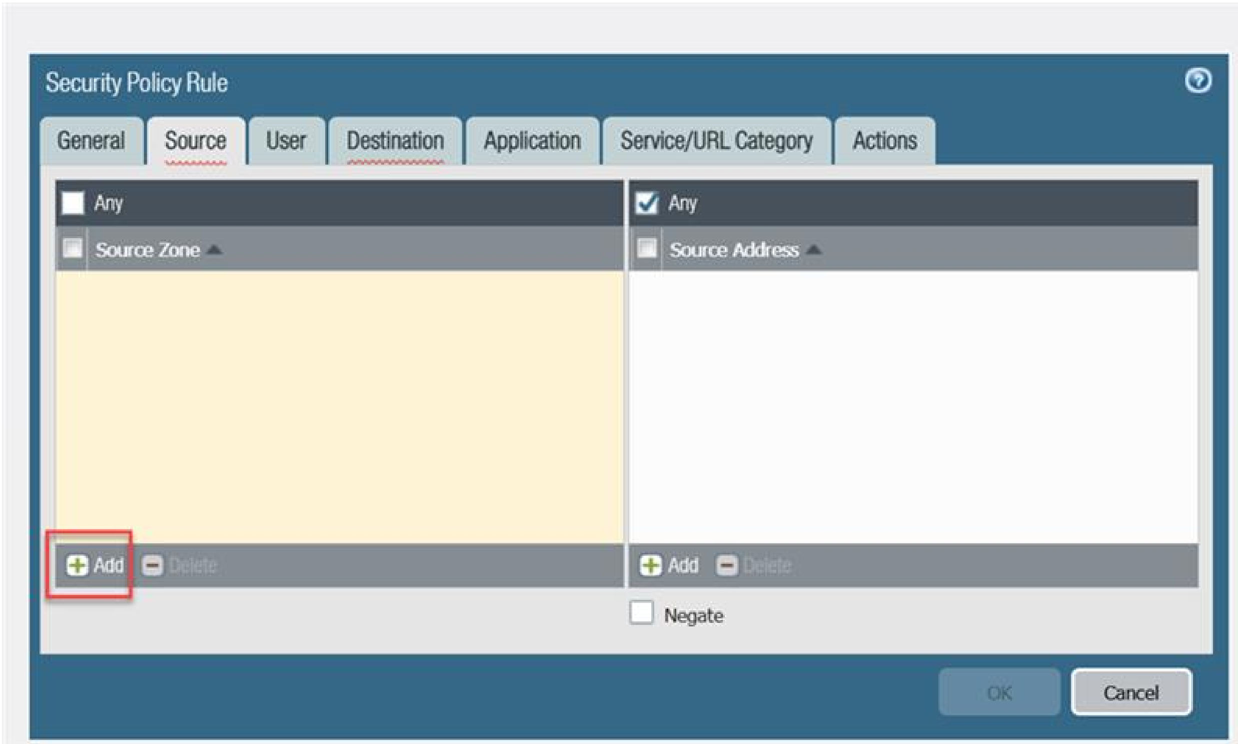
5. Click the Policies tab on the top of the GUI as highlighted in the image above.



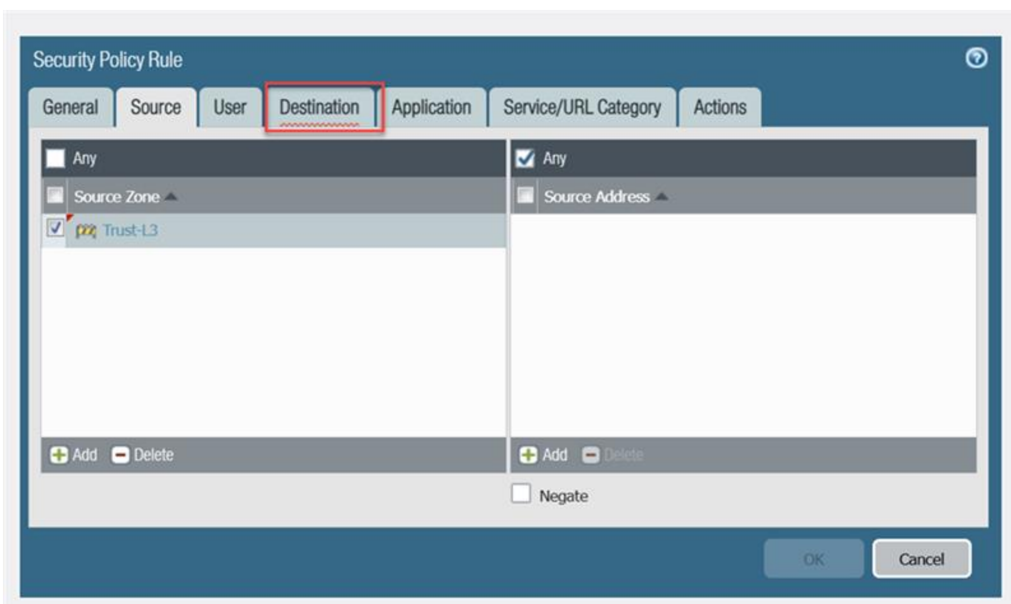
5. Click the Add button as highlighted in the image above.



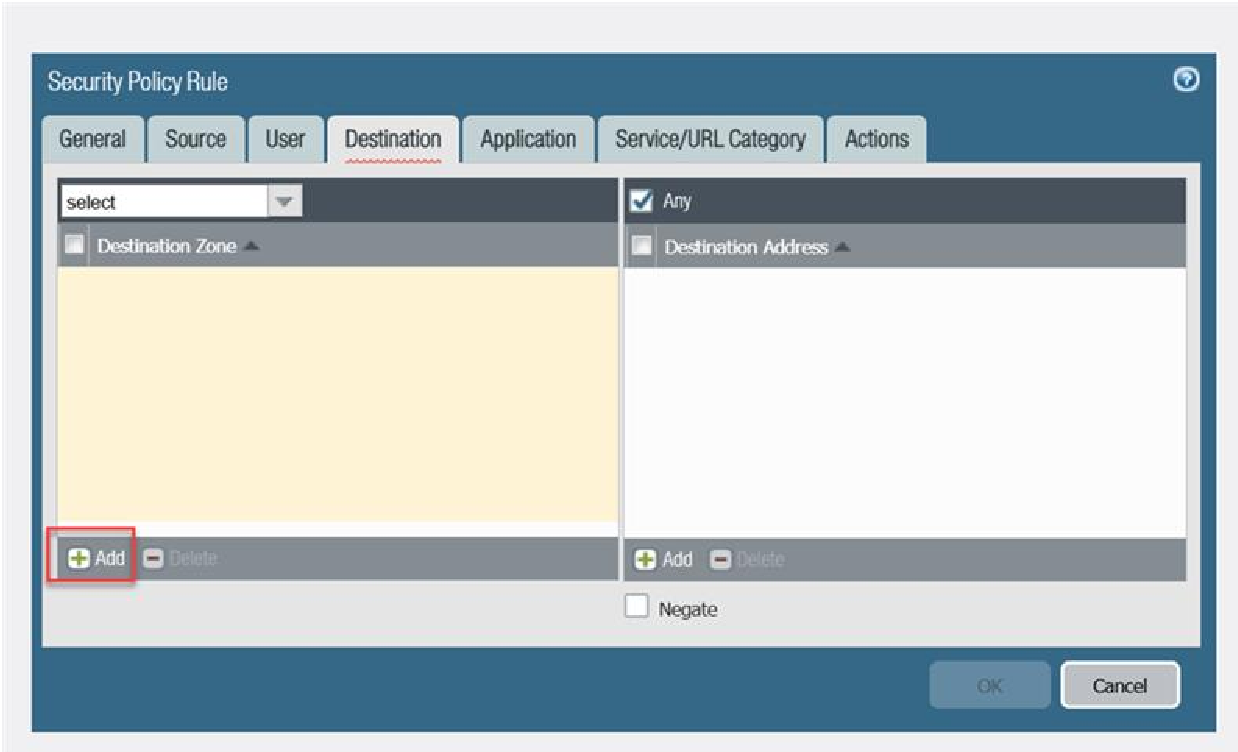
5. Modify the settings as highlighted in the image above. Then, select the Source tab as highlighted in the image above.



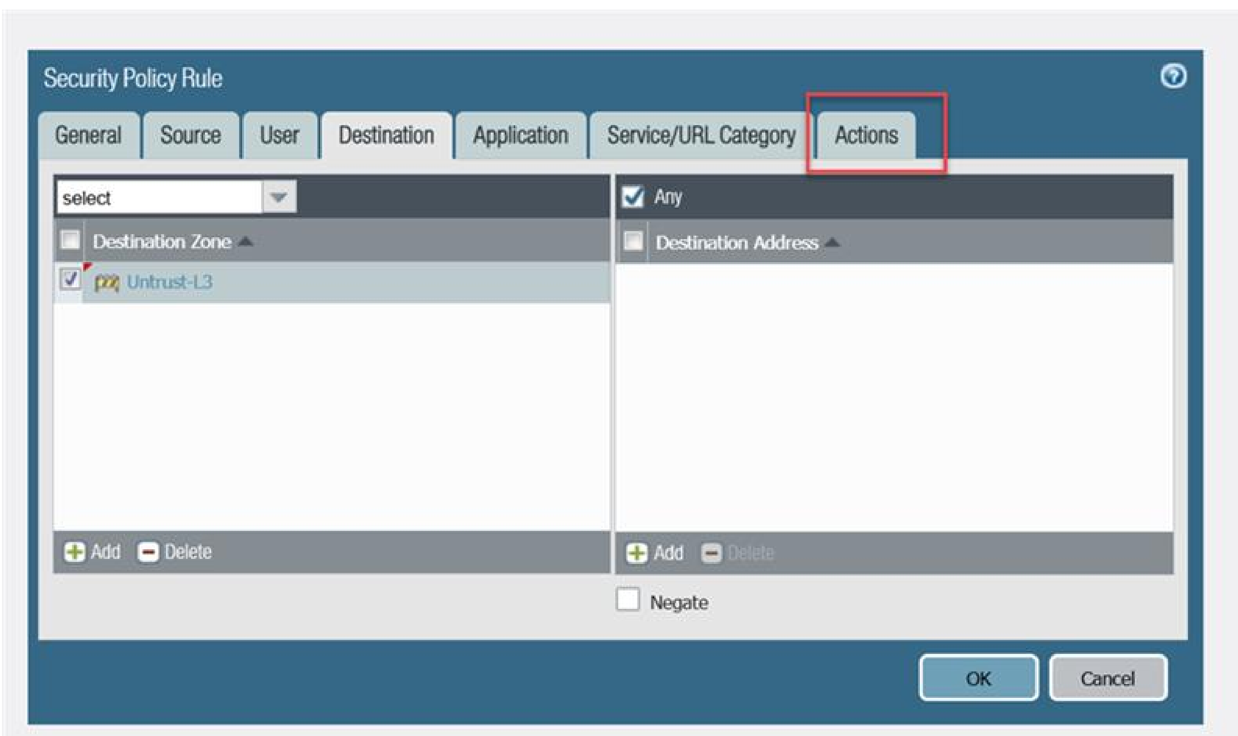
5. Select the Add button as highlighted in the image above. Then type in Trust-L3 before pressing the enter button.



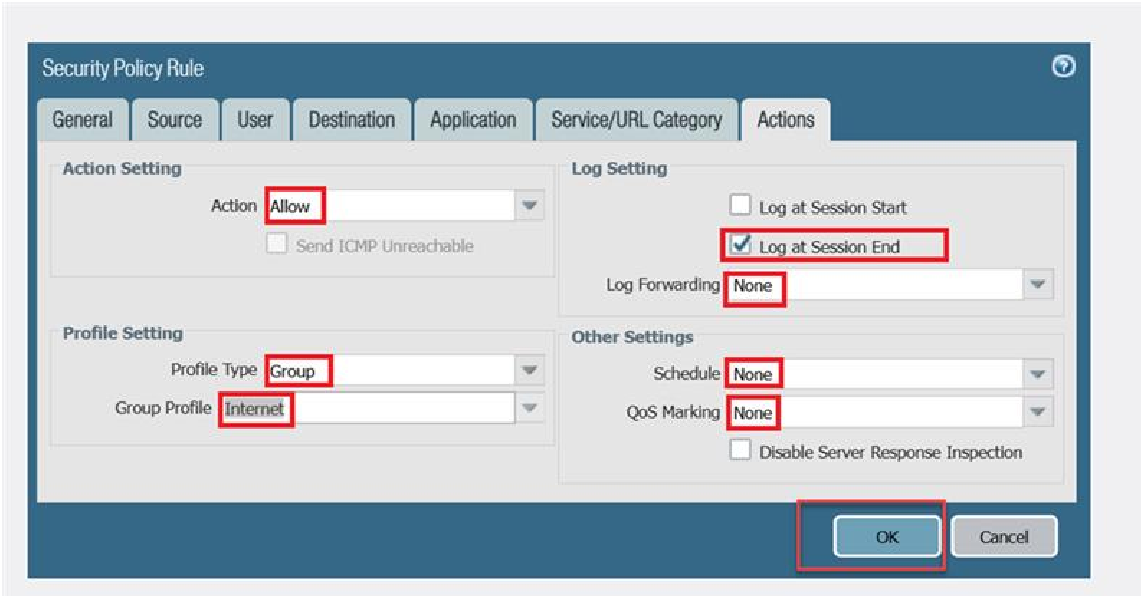
5. Click on the Destination tab as highlighted in the image above.



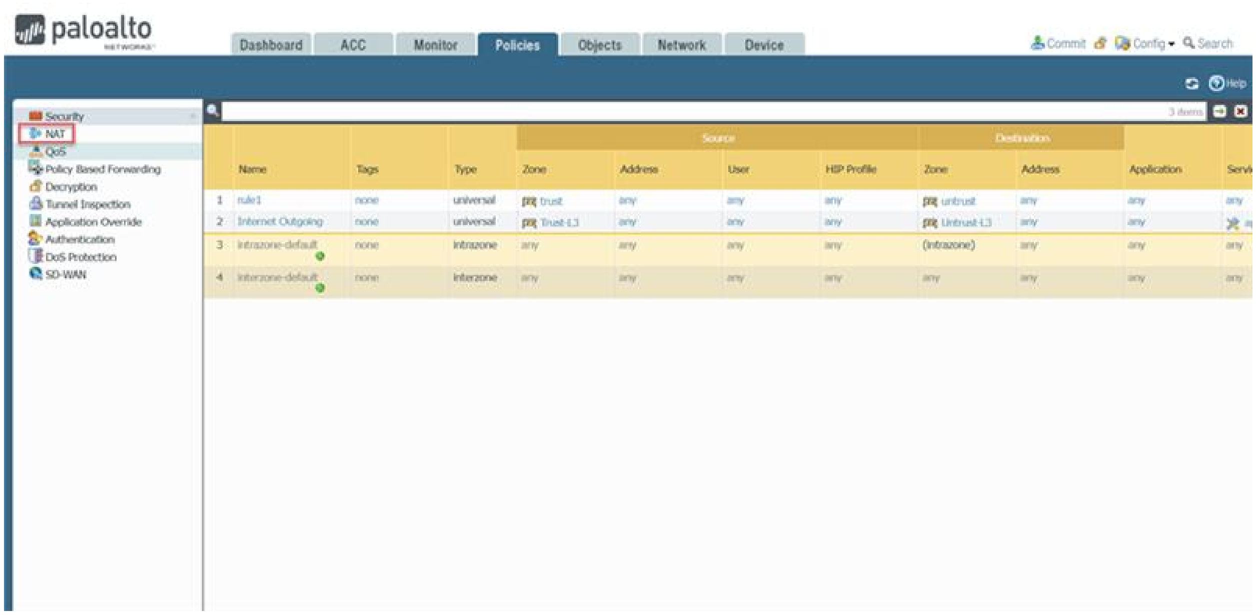
5. Select the Add button as highlighted in the image above. Then type in Untrust-L3 before pressing the enter button.



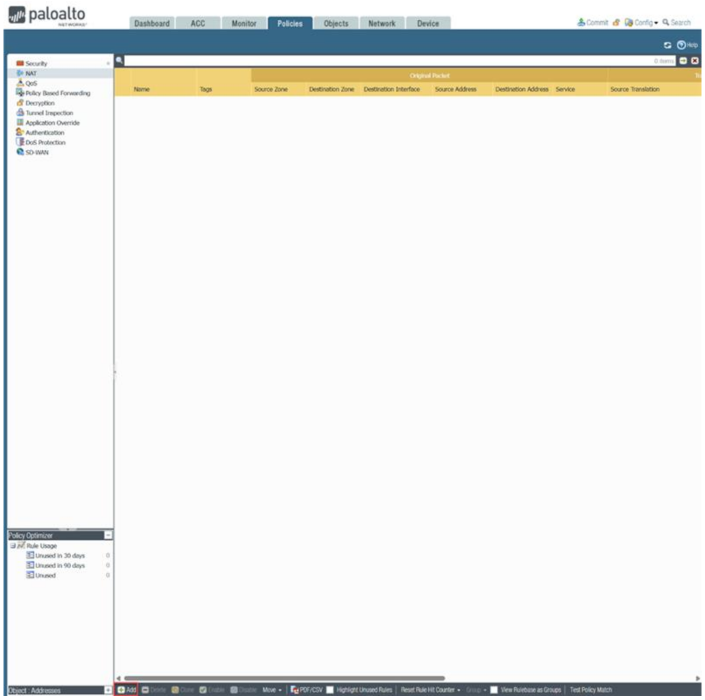
5. Select the Actions tab as highlighted in the image above.



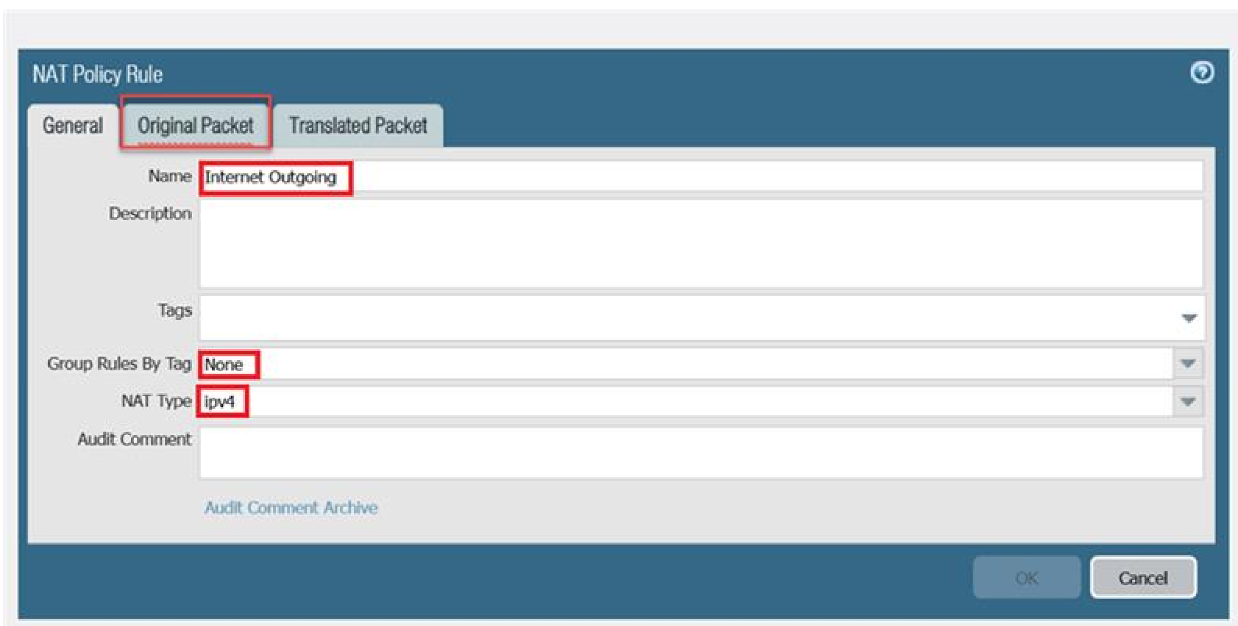
5. Modify the settings as highlighted in the image above. Then, select the OK button.



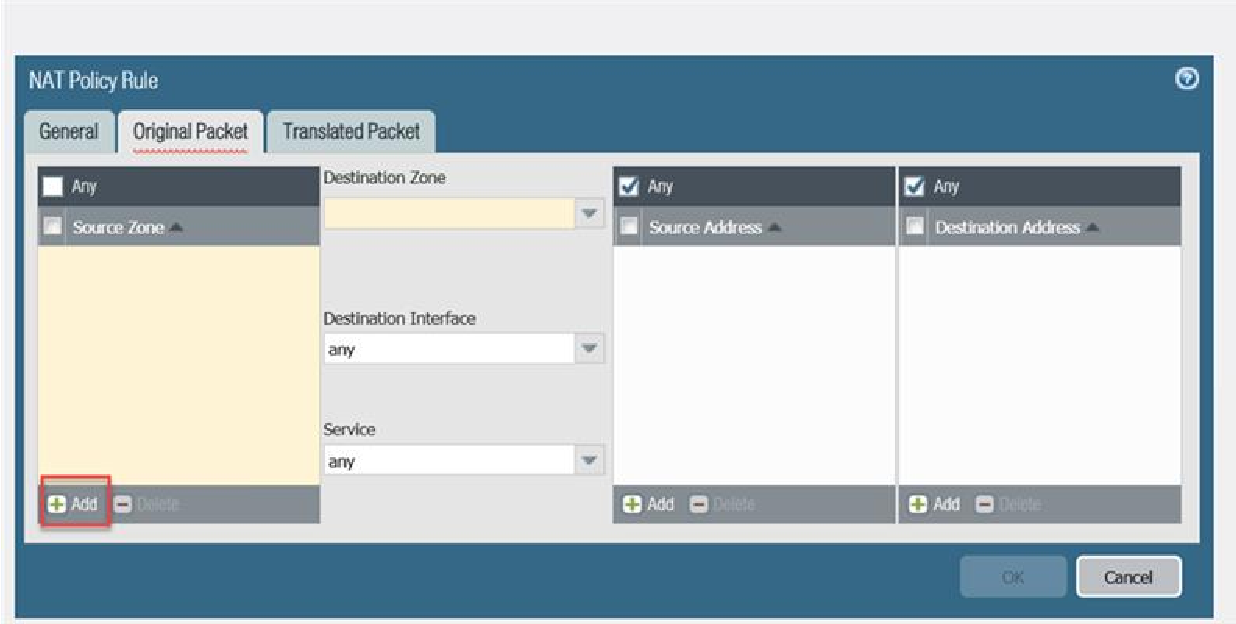
5. Select the NAT section on the left of the GUI as highlighted in the image above.



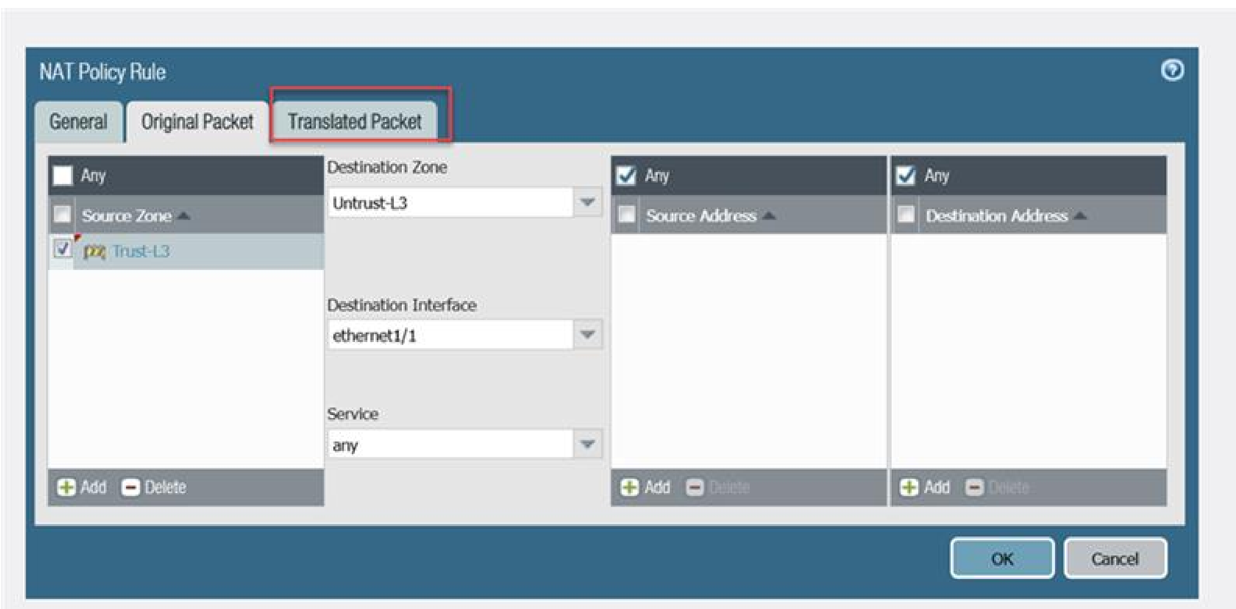
5. Select the Add button as highlighted in the image above.



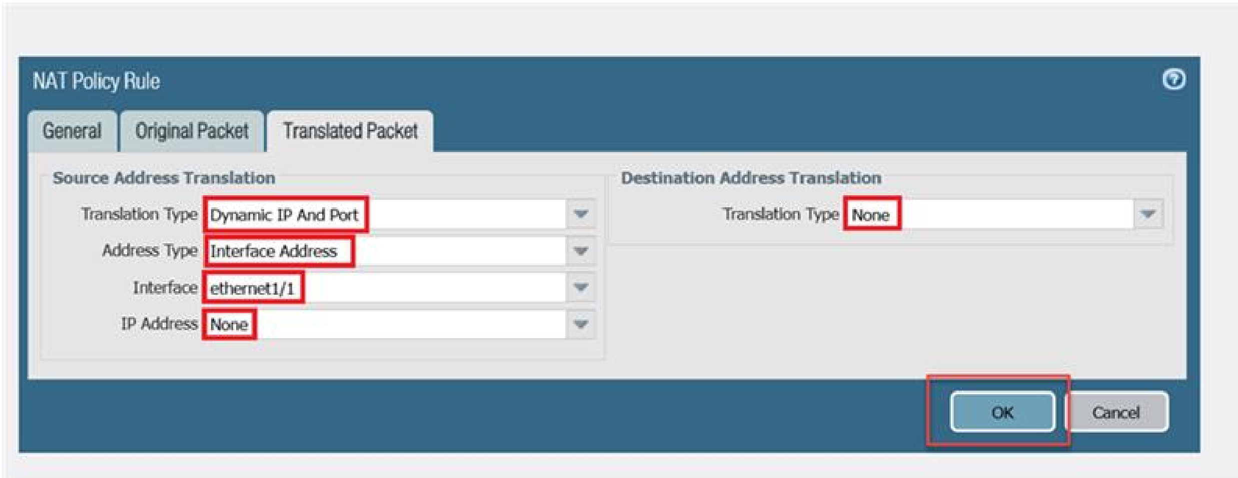
5. Modify the settings as highlighted in the image above. Then, select the Original Packet, as highlighted in the image above.



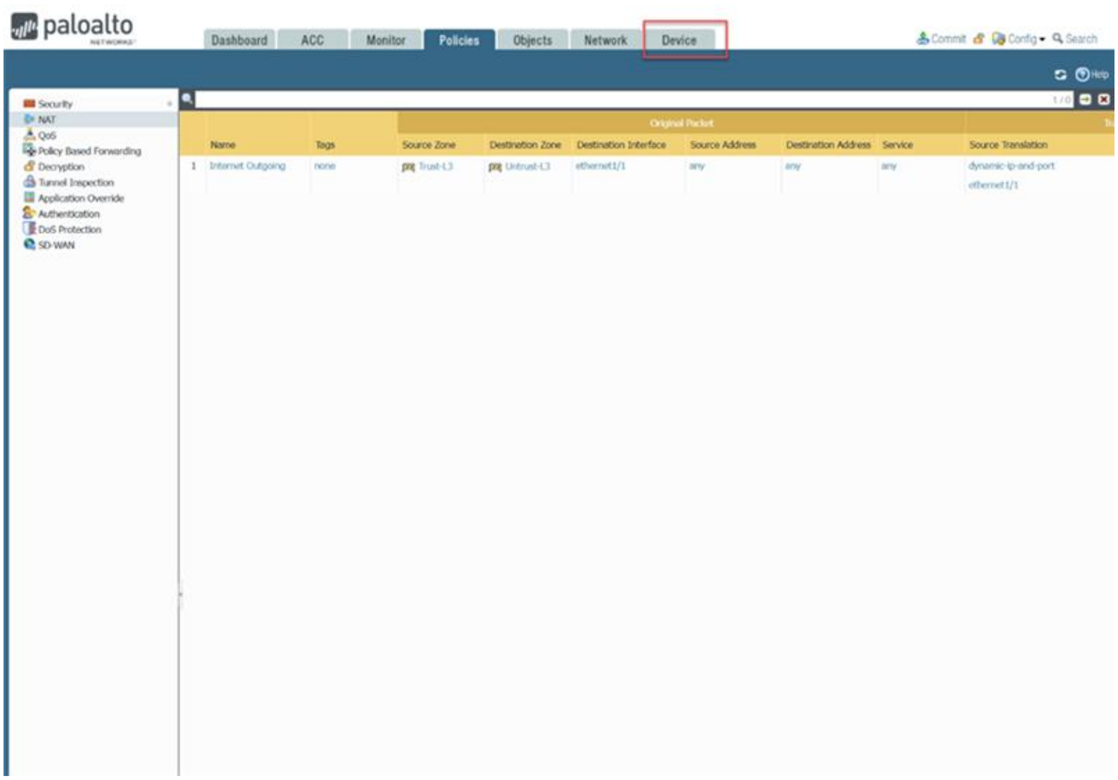
5. Click the Add button as highlighted in the image above. Type Trust-L3 before pressing the enter key.



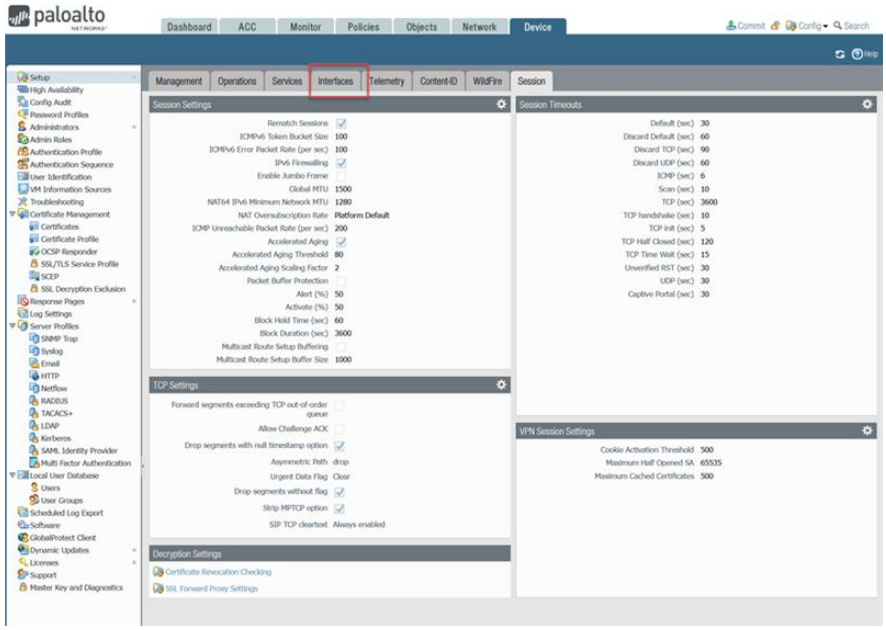
5. Select the Translated Packet tab as highlighted in the image above.



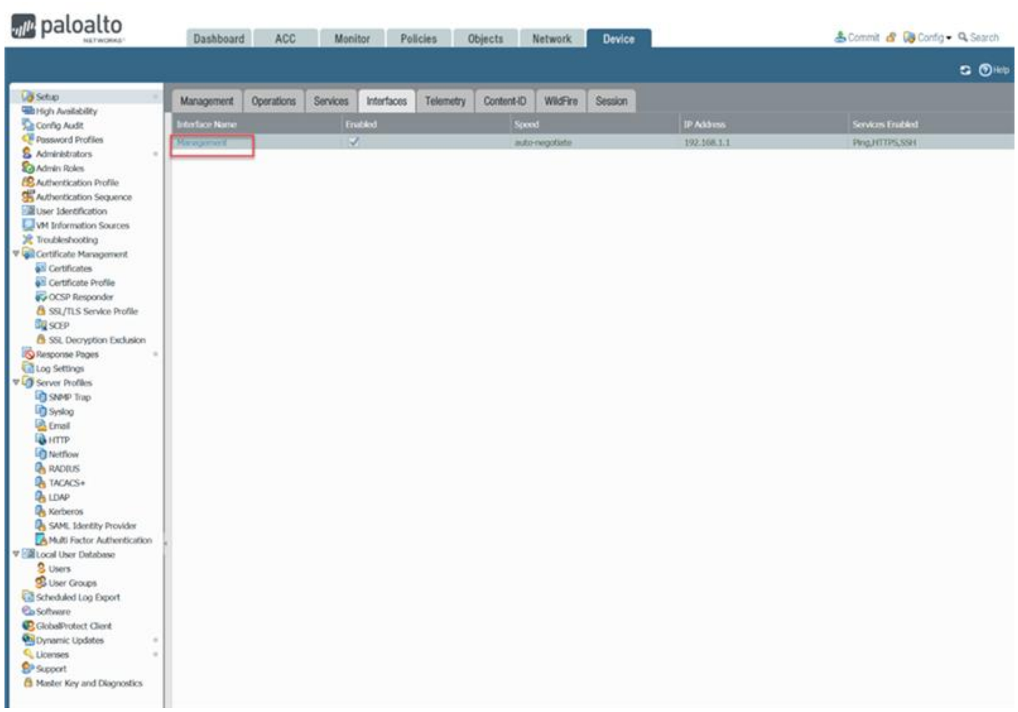
5. Modify the settings as highlighted in the image above. Then, select the OK button.



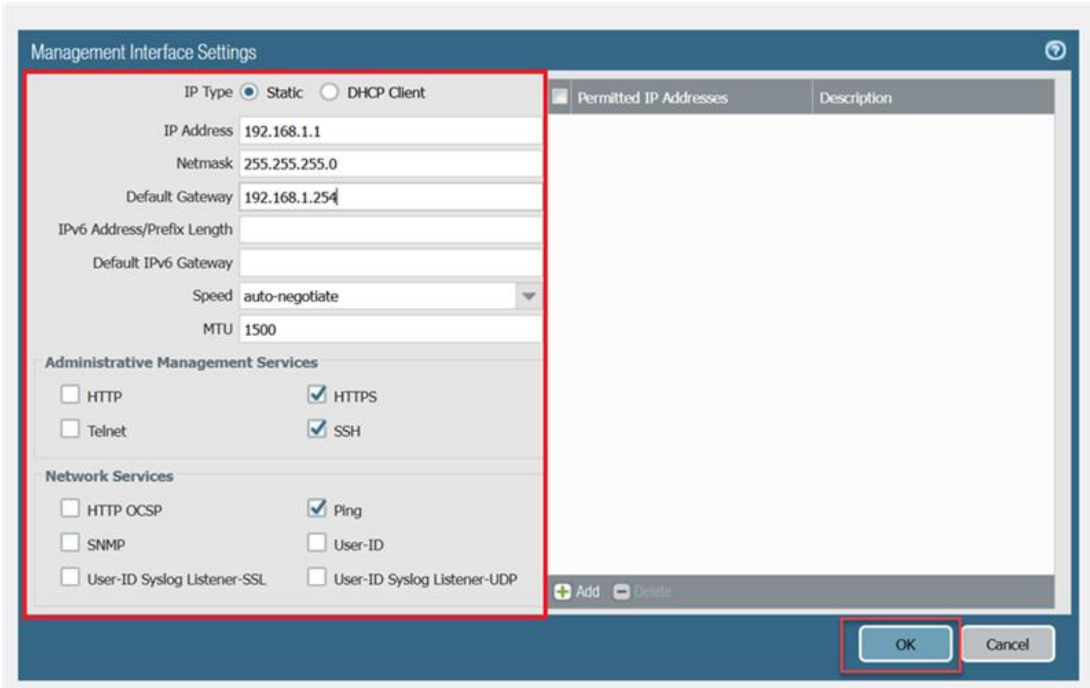
5. Select the Device tab on the top of the GUI as highlighted in the image above.



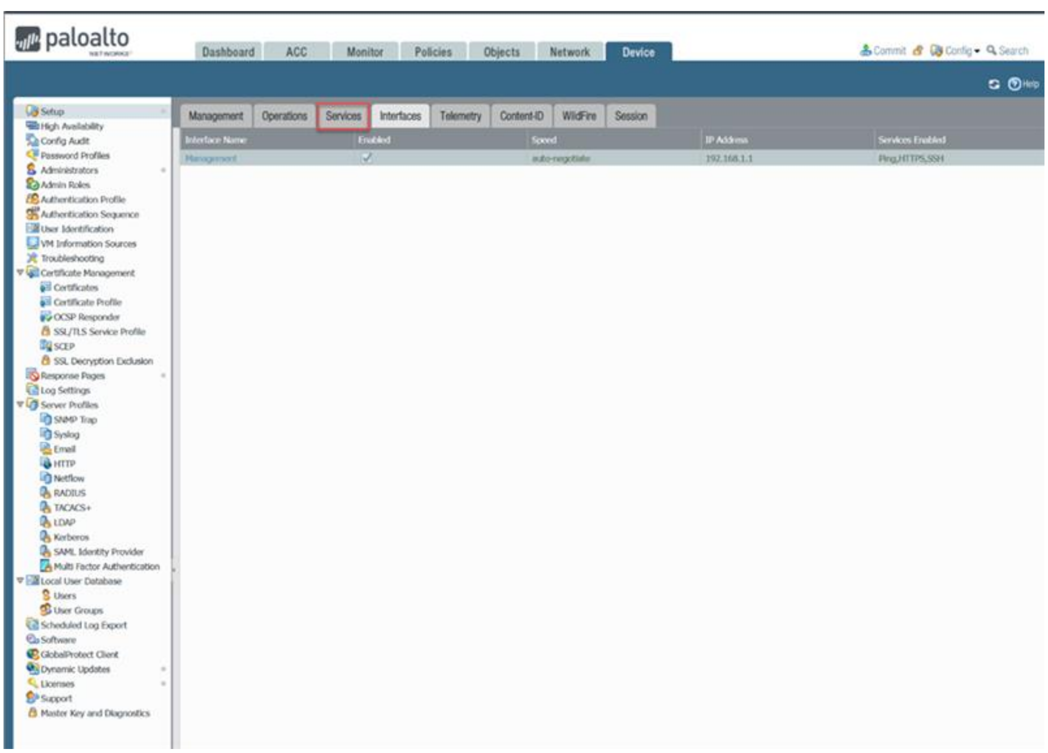
5. Check if all configurations in the Session tab are matching as in the image above. Then, select the Interfaces tab as highlighted in the image above.



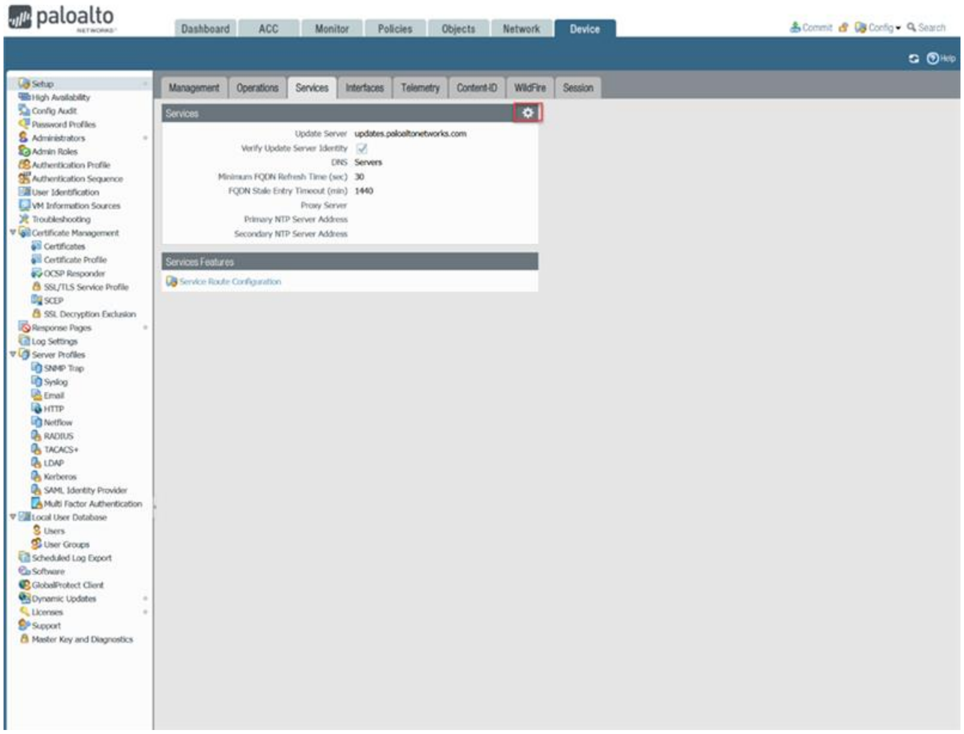
5. Select the Management Interface as highlighted in the image above.



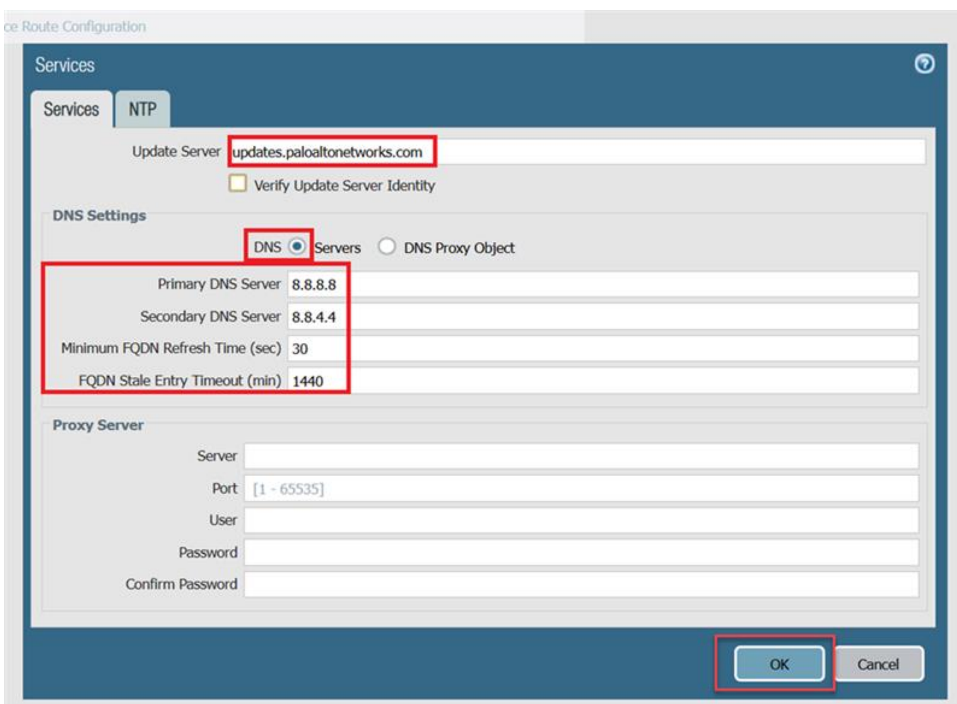
5. Modify the settings as highlighted in the image above. Then, select the OK button as highlighted in the image above.



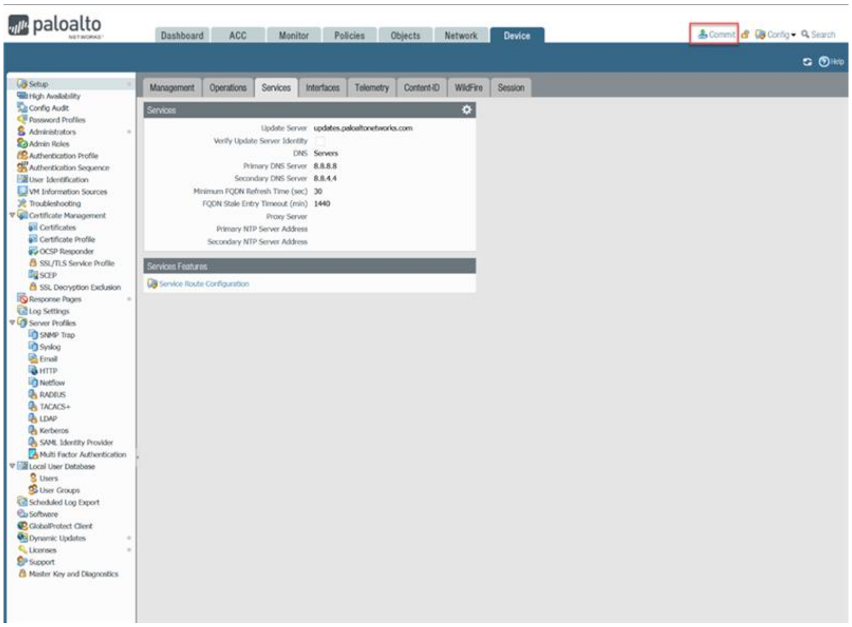
5. Select the Services tab as highlighted in the image above.



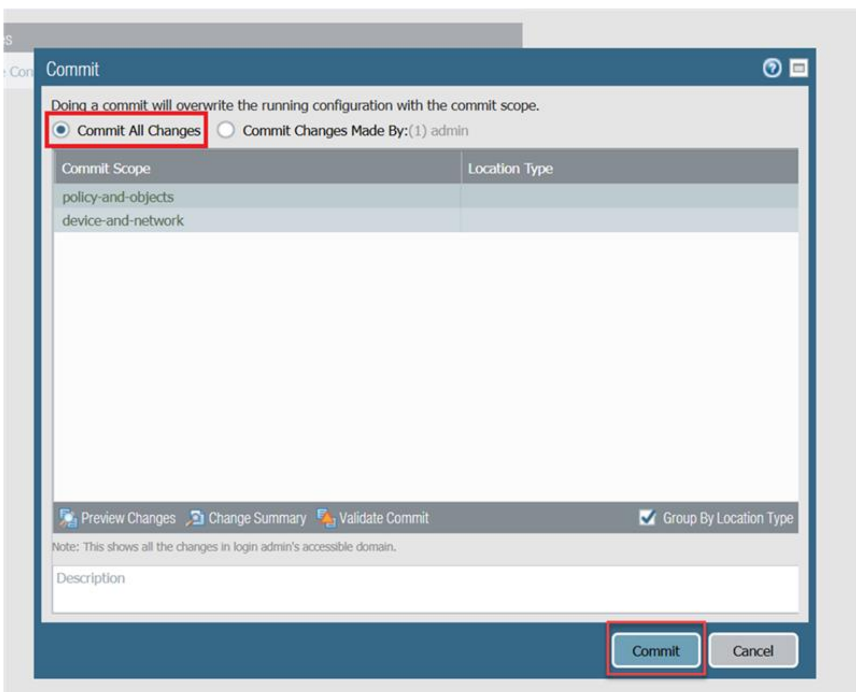
5. Click on the System icon as highlighted in the image above.



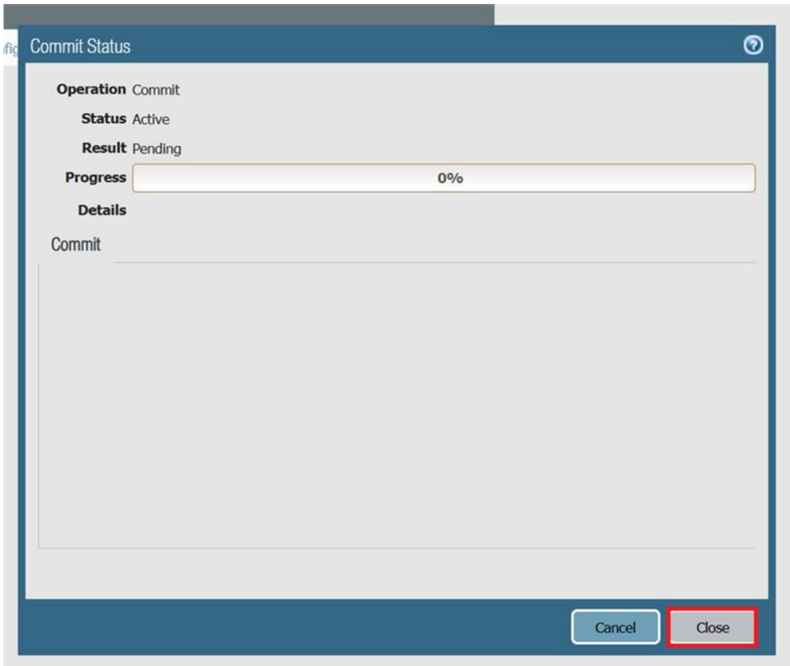
5. Modify the settings as highlighted in the image above. Then, select the OK button.



5. Press the Commit button as highlighted in the image above in the top right corner of the GUI.



5. Select the Commit All Changes option as highlighted in the image above. Then, select the Commit button.



5. When the Progress of the Commit reaches 100% press the Close button as highlighted in the image above

Problems

We had to update Pan-OS from 9.1 to 10.2.9, so we had to install and download each version, which took 30 minutes. When we got the new version, we lost the device certificate, so we had to generate a one-time password and obtain a certificate from Palo-Alto which delayed the completion of this lab. We also had to re-configure the SOHO on the new OS.

Our WAN port leading to internet switch had layer-1 problems relating to the ethernet cable.

Conclusion

The PA220's web interface was confusing to navigate at first, but after configuring the firewall for a SOHO network, I believe I have gained a strong understanding of how to navigate PAN-OS's interface. All in all, a SOHO network configuration using a PA220 firewall had a relatively simple setup process and would work great for an individual or small business

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